

ADVERSARIAL ATTACKS LEVERAGE INTERFERENCE BETWEEN FEATURES IN SUPERPOSITION

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ABSTRACT

Fundamental questions remain about when and why adversarial examples (AExs) arise in neural networks (NNs), with competing views characterising them either as artifacts of the irregularities in the decision landscape or as products of sensitivity to non-robust input features. In this paper, we instead argue that adversarial vulnerability can stem from *efficient* information encoding in NNs. Specifically, we show how superposition – where networks represent more features than they have dimensions – creates arrangements of latent representations that adversaries can exploit. Across precisely controlled synthetic settings, we find that gradient-based (projected gradient descent (PGD)) attacks align almost perfectly with the theoretically optimal interference-exploiting perturbation (cosine similarity 0.92–0.97 versus 0.00 for a matched-norm random baseline, $p < 10^{-10}$), establishing that adversarial perturbations systematically leverage interference between superposed features rather than perturbing inputs arbitrarily. This mechanistic account makes attack patterns quantitatively predictable from feature arrangements and yields a direct, measurable explanation for two known phenomena: attack transferability between models with similar training regimes and class-specific vulnerability patterns. Strengthening input correlations, for instance, raises cross-model attack transfer from 18% (uncorrelated) to 94% (globally correlated) in lockstep with rising geometric consistency between independently trained models. In these synthetic settings with precisely controlled superposition, we establish that superposition *suffices* to create adversarial vulnerability. We then show that every one of these quantitative trends persists in a vision transformer (ViT) trained on CIFAR-10: as the bottleneck is compressed from $m = 10$ to $m = 2$, robust accuracy falls and transferability rises monotonically, mirroring the toy-model pattern. Together, these findings reveal adversarial vulnerability can be a byproduct of networks’ representational compression, rather than flaws in the learning process or non-robust inputs. We release our implementation through a software repository.

1 INTRODUCTION

Despite extensive research on AExs (Szegedy et al., 2014; Goodfellow et al., 2014), there remains no consensus on their fundamental cause. We still lack a complete explanation for how minimal perturbations so drastically alter model predictions, leaving us unable to predict which perturbations succeed, quantify why attacks transfer between models, or design principled defences.

Existing explanations broadly fit into two camps (Nakkiran, 2019). The *bug* perspective attributes vulnerability to factors including learning flaws (Schmidt et al., 2018; Nakkiran, 2019) or geometric properties of high-dimensional decision boundaries (Fawzi et al., 2016). Work in this domain focuses on providing statistical bounds on properties such as capacity (Bubeck & Sellke, 2021) or Lipschitz constraints (Hein & Andriushchenko, 2017) without explaining which specific perturbations succeed. The *feature* perspective argues AExs exploit predictive but non-robust statistical patterns in data (Ilyas et al., 2019), reframing the debate to how underlying data relates to human perception. It treats these non-robust features as fixed properties of the data rather than arising from network encoding strategies. Neither approach reconciles how representational constraints interact with data semantics.

In this paper, we bridge this divide through a mechanistic interpretability perspective, demonstrating how adversarial vulnerability emerges from the interaction between architectural constraints and data semantics (*i.e.* human interpretable properties of the data). We show that AExs can emerge from interference between learned representations in NNs. Adversarial attacks systematically exploit

superposed latent structure – a mechanism that normally enables additional representation capacity – to craft effective perturbations that manipulate model outputs. Crucially, this mechanism is not merely qualitative: PGD-discovered perturbations coincide with our theoretically derived interference-exploiting optimum at cosine similarities of 0.92–0.97, while a matched-norm random baseline sits at 0.00. Our analysis reveals a mechanistic pathway: input correlations constrain feature arrangements, these arrangements determine interference patterns, and these interference patterns dictate attack characteristics and transferability. Each link in this chain is quantitatively verifiable – stronger correlations raise geometric consistency, which in turn raises attack transfer from 18% to 94% – so the framework predicts not only *which* perturbations succeed but *how often* they transfer between models.

Our account of AExs draws on recent theories about how networks encode and process information, namely the linear representation hypothesis (LRH) (Park et al., 2024) and the theory of superposition (Elhage et al., 2022). The LRH posits that *input features*—fundamental abstractions of data—are represented as linear directions in a network’s representation space. It is hypothesised that NNs can represent significantly more of these features than they have neurons through superposition, enabling networks to efficiently pack multiple features into shared dimensions at the cost of introducing interference. Such interference means perturbing one feature can affect others in non-obvious ways. This paper investigates whether this interference creates vulnerabilities that AExs exploit, and what insights this offers for understanding adversarial phenomena. Our primary contributions are:

- We present a framework reconciling architectural constraints with feature properties in adversarial vulnerability. Using simple models, we demonstrate how data properties induce specific superposition geometries that directly determine realised adversarial perturbations. We quantify this correspondence directly: PGD attacks match the optimal interference-exploiting direction at 0.92–0.97 cosine similarity ($p < 10^{-10}$ against a 0.00 random baseline), and shared geometry predicts transfer rates that span 18% to 94% as input correlations strengthen.
- We demonstrate these findings persist in a ViT trained on CIFAR-10 with an engineered bottleneck layer to induce superposition, where compressing the bottleneck from $m = 10$ to $m = 2$ monotonically lowers robust accuracy and raises transferability, reproducing the toy-model trends.
- We show superposition to be sufficient, but not necessary, for adversarial vulnerability. In identifying this, we establish algorithmic brittleness as a distinct vulnerability mechanism.
- We demonstrate that mechanistic understanding of learned representations enables informed adversarial attack construction: a gradient-free, representation-informed perturbation succeeds at ℓ_2 budget 0.22, within $1.3\times$ of PGD (0.17) and far below naive baselines (up to 6.20), underpinning our argument for semantically-informed defences.

2 BACKGROUND

We now provide an overview of the tools we use in our analysis: the LRH and superposition. Formally, let $\mathbf{x} \in \mathcal{X}$ denote the input to a NN and $\mathbf{h}^{(l)} \in \mathbb{R}^{d_l}$ the activation vector of the l -th layer.

Linear representation hypothesis (LRH). The LRH posits that NNs represent many variables of their computation, such as semantic properties of their inputs, as linear directions in their activation space, which can be used as abstractions for reasoning (Park et al., 2024; Guerner et al., 2023). This bias towards representing linear features is hypothesised because linear separability allows networks to easily recognise and manipulate features, and because dot products with subsequent layer weights efficiently process such directional features. A growing body of empirical work supports this account (Gurnee et al., 2023; Park et al., 2024). Let $\mathcal{C} = \{c_1, \dots, c_M\}$ denote a set of M semantically meaningful latent features (*e.g.* concepts like “presence of shape,” or “indoor vs. outdoor”). Formally:

Definition 1 (Linear Representation Hypothesis (LRH)) A neural network layer with activations $\mathbf{h}^{(l)} \in \mathbb{R}^{d_l}$ satisfies the LRH if it represents the latent features $\mathcal{C} = \{c_1, \dots, c_M\}$ as linear directions $\{\mathbf{v}_j\}_{j=1}^M \subset \mathbb{R}^{d_l}$ such that:

$$\mathbf{h}^{(l)}(\mathbf{x}) \approx \sum_{j=1}^M a_j(\mathbf{x}) \mathbf{v}_j$$

where $a_j(\mathbf{x}) \in \mathbb{R}$ represents the activation magnitude of feature c_j , and $\mathbf{v}_j$ is the corresponding linear direction. The features are **linearly accessible** (Costa et al., 2025): inputs $\mathbf{x}_0, \mathbf{x}_1 \in \mathcal{X}$ that

differ mainly in the value of feature c_j while holding other features approximately constant (i.e., $a_j(\mathbf{x}_1) - a_j(\mathbf{x}_0) = k$ and $|a_i(\mathbf{x}_1) - a_i(\mathbf{x}_0)| < \lambda$ for all $i \neq j$ and some small $\lambda > 0$) satisfy:

$$\mathbf{h}^{(l)}(\mathbf{x}_1) - \mathbf{h}^{(l)}(\mathbf{x}_0) \approx k\mathbf{v}_j$$

where k reflects the change in c_j .

Superposition and sparsity. NNs are capable of representing many more latent features than there are available dimensions in activation space: $M > d_l$. For example, LLMs can reference many more place names than they have residual stream dimensions. One framework for analysing this phenomenon is superposition, according to which networks use an overcomplete and non-orthogonal set of feature directions $\{\mathbf{v}_j\}_{j=1}^M$. This leverages the fact that $2^{\Theta(d\epsilon^2)}$ almost orthogonal vectors ($< \epsilon$ cosine similarity) can be represented in d -dimensional space (Tkocz, 2012) – an exponential capacity gain – and that sparse vectors can be accurately recovered after projection into lower-dimensional spaces (Elhage et al., 2022; Bereska & Gavves, 2024; Sawmya et al., 2025). This efficiency comes at the cost of two challenges. First, *polysemanticity* (Scherlis et al., 2022; Lecomte et al., 2023) emerges where individual neurons contribute to multiple different features, meaning a neuron’s activation does not correspond to a single interpretable concept. Second, non-orthogonal feature directions create *interference* between features – activating one feature activates others. Networks can mitigate these issues through non-linear operations (e.g. ReLU, softmax) that disambiguate superposed features (Gurnee et al., 2023).

Definition 2 (Superposition Hypothesis) A network layer represents features in superposition if:

1. The number of latent features exceeds layer dimensionality: $M > d_l$
2. There exists non-orthogonal feature directions: $\exists i, j$ with $i \neq j$ such that $\mathbf{v}_i \cdot \mathbf{v}_j \neq 0$
3. Latent feature activations are sparse: $\mathbb{E}_{\mathbf{x}}[\|\mathbf{a}(\mathbf{x})\|_0] \ll M$, where $\mathbf{a}(\mathbf{x}) = [a_1(\mathbf{x}), \dots, a_M(\mathbf{x})]^T$

The network trades representational capacity against feature interference by packing more features than dimensions.

Adversarial attacks. Next we define the notion of adversarial perturbations and attacks following (Szegedy et al., 2014; Costa et al., 2024):

Definition 3 (Adversarial attack) Let $\mathbb{B}_p(\mathbf{x}, \epsilon)$ denote the ϵ -ball around $\mathbf{x} \in \mathcal{X}$ in the ℓ_p -sense, e.g. $\mathbb{B}_\infty$ is the ℓ_∞ -ball. Any small δ that p -perturbs $\mathbf{x}$ to create an **adversarial input** $\mathbf{x}' \in \mathbb{B}_p(\mathbf{x}, \epsilon)$ such that $f(\mathbf{x}') \neq f(\mathbf{x})$ (untargeted) or $f(\mathbf{x}') = y_{\text{target}}$ (targeted) is known to be an **adversarial attack**. ℓ_∞ - or ℓ_2 -norm attacks are created via the PGD (detailed in Sec. A).

Definition 4 (Adversarial robustness) A classifier $g_w(\cdot)$ with parameters w is said to be **robust** at $\mathbf{x}$ with radius $\epsilon > 0$, if $g_w(\mathbf{x}') = g_w(\mathbf{x}) \forall \mathbf{x}' \in \mathbb{B}(\mathbf{x}, \epsilon)$.

3 SUPERPOSITION GEOMETRY CAN DETERMINE ADVERSARIAL ATTACKS

We begin by investigating if and how adversarial attacks exploit the interference rooted in the superposition of latent features. Specifically, we ask three questions:

1. Do adversarial perturbations exploit the interference between superposed features?
2. Do correlations in the input shape the geometric arrangement of superposed latent features?
3. Can shared latent geometry explain why attacks transfer between independently trained models?

Experimental setup. To investigate these questions, we design a synthetic task that exemplifies established LRH and superposition phenomena (Park et al., 2024; Elhage et al., 2022) in a precisely controllable setting. The task is designed to: (1) provide an intuitive classification setting for studying AExs; (2) represent class concepts as linear directions per the LRH; (3) induce controlled superposition between these latent features; and (4) retain a priori knowledge of how inputs correspond to the superposed features — enabling testable predictions about adversarial mechanisms.

Specifically, we partition input $\mathbf{x} \in \mathbb{R}^d$ into k equally-sized groups $\mathbf{x} = [\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)}]$, where $\mathbf{x}^{(j)}$ represents the p input features associated with class j . The task identifies which group has the largest sum: $y = \operatorname{argmax}_{j \in \{1, \dots, k\}} \sum_{i=1}^p x_i^{(j)}$. Intuitively, each $x_i^{(j)}$ represents evidence for class j ,

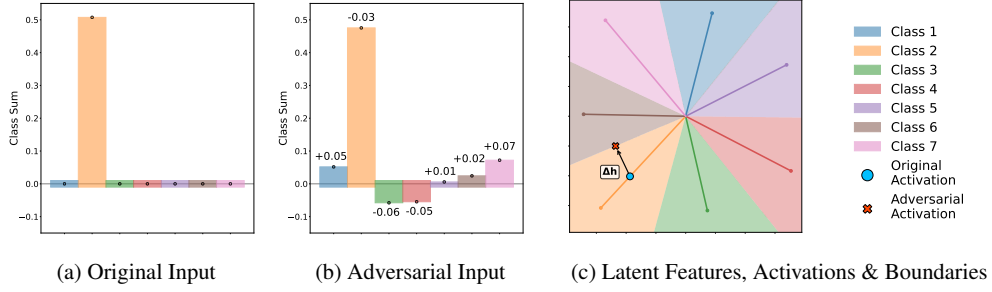


Figure 1: **An adversarial attack exploiting superposition geometry** ($k = 7, m = 2$). (a) The original sample. (b) The adversarially perturbed sample: its ground truth is unchanged yet it is misclassified, and the sign and magnitude of every input perturbation are fixed by the configuration of latent representations rather than chosen arbitrarily. (c) The same original and adversarial sample in activation space, where the arrows are the column vectors of $\mathbf{W}_e$ (the latent representations of the input features); the perturbation in (b) follows directly from the dot products between these directions and the attack vector.

with the p features capturing different class attributes. Following Elhage et al. (2022), we sample $x_i^{(j)} \sim \text{Uniform}(0, 1)$ with sparsity S (probability of setting to zero).

A two-layer network compresses the input into m dimensions through a bottleneck encoder, $\mathbf{h} = \text{ReLU}(\mathbf{W}_e \mathbf{x} + \mathbf{b}_e) \in \mathbb{R}^m$ (where $m < k < d$), and reconstructs them via a decoder, $\mathbf{z} = \mathbf{W}_d \mathbf{h} + \mathbf{b}_d \in \mathbb{R}^k$. Our primary setup uses cross entropy (CE) loss without ReLUs or biases. This is chosen for its simplicity, with the softmax limiting interference by exponentially suppressing contributions from interfering latent features. To ensure our findings are not specific to this context, we validate our findings using a second setup with mean squared error (MSE) loss with ReLUs and biases. The representational challenge is to compress information from $d = k \times p$ dimensions to $m < k$ dimensions, and then recover sufficient information to identify the class with the largest sum.

While non-synthetic tasks will have features representing nuanced concepts that require discovery and interpretation, this setup provides us with ground truth class labels for the latent features. This allows us to compute the true class after any perturbation by recalculating group sums, facilitating our definition of an AEx as satisfying the following two conditions:

1. The model’s prediction under perturbation changes (e.g. from class A to B)
2. The true class remains unchanged (e.g. still class A)

This distinguishes genuine AExs from perturbations that actually change the ground truth (i.e. change which group has the highest sum). Attacks are created using ℓ_∞ - and ℓ_2 -norm PGD.

Evaluation metrics. We interpret the columns of the encoder weight matrix $\mathbf{W}_e$ as the latent representations $\mathbf{v}_i$ for each input feature x_i . Since input feature $x_i^{(j)}$ contributes exclusively to class j , these $\mathbf{v}_i$ (for $i \in \text{class } j$) align. The direct correspondence between input features and class-aligned latent directions allows us to precisely trace how perturbations to the inputs of a specific class, $\mathbf{x}^{(j)}$, relate to changes in latent space. For successful PGD-generated AExs $\mathbf{x}_{adv} = \mathbf{x}_{orig} + \delta$, we measure:

1. **Input perturbation profile (IPP):** The sign and magnitude of each δ_i , describing how different input features are perturbed in an attack.
2. **Latent attack alignment:** The similarity $\Delta \mathbf{h} \cdot \mathbf{v}_c$ between the *latent attack vector* $\Delta \mathbf{h} = \mathbf{h}_{adv} - \mathbf{h}_{orig}$ and class c ’s representation.
3. **Attack transferability:** Success rate of attacks generated on one model when applied to another.
4. **Robust accuracy:** The fraction of examples that remain correctly classified after perturbation δ with $\|\delta\|_p \leq \epsilon$ to a sample of inputs.

3.1 RESULTS

Do attacks exploit interference between latent features in superposition? An attack must move a sample across a decision boundary to change its class, but what determines the required input perturbations? Two contrasting intuitions exist: the *feature intuition* suggests increasing input

features of the target class, while the *bug intuition* suggests adversarial perturbations modify inputs in unpredictable ways, with knowledge of feature representations providing no insight.

Fig. 1 shows a typical AEx in our setting, showing an input of Class 2 (orange) and its perturbed value (with IPP) that misclassifies it as Class 6 (brown). At first glance the perturbations appear arbitrary – we clearly do not simply increase the target class features, ruling out the feature intuition. Yet they are equally clearly not random: the sign and magnitude of δ track the configuration of latent features with a precision we quantify below. This systematic correspondence points to a third intuition: *attacks are mediated by latent geometry*.

Quantitatively, the IPP tracks the latent attack alignment closely. The sign and magnitude of each per-class perturbation $\delta^{(j)}$ move in step with how that class’s latent representation $\mathbf{v}^{(j)}$ aligns with the latent attack vector $\Delta\mathbf{h}$: positively aligned representations ($\mathbf{v}^{(j)} \cdot \Delta\mathbf{h} > 0$) have their input features proportionally amplified, while negatively aligned representations are attenuated. This is exactly what Fig. 1(b) illustrates – input perturbations set by the dot products in Fig. 1(c) to push the sample across the decision boundary.

To test directly whether PGD exploits superposition geometry, we compare PGD-generated attacks against the theoretically optimal interference-exploiting perturbation (derived in Sec. 3.2) and against a matched-norm random baseline. For each configuration (k, m) , we train 5 models under different random seeds and generate 1000 PGD attacks per model, measuring the cosine similarity between each successful perturbation and the corresponding optimum (samples with ℓ_2 norm below ϵ are filtered out to ensure meaningful perturbations). The comparison is decisive: as Tab. 1 reports, PGD aligns with the optimum at 0.97 ($k=6, m=2$), 0.96 ($k=30, m=10$), and 0.92 ($k=90, m=30$), whereas the random baseline never rises above 0.00 ± 0.02 . PGD thus recovers essentially the entire optimal direction while a norm-matched random perturbation recovers none of it, and the gap holds across a $15\times$ increase in the number of classes. One-sample t-tests against the random baseline give $p < 10^{-10}$ for every configuration, so this near-perfect match is not chance. Because the optimal attack provably leverages interference and PGD reproduces it almost exactly, we conclude that PGD exploits superposition interference.

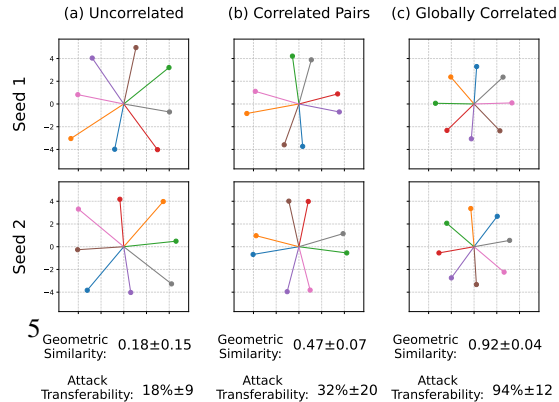
Finding: Adversarial attacks systematically exploit interference between superposed features. Successful PGD attacks are predictable given the specific superposition geometry, rather than random.

Do input correlations determine latent feature geometry? We now investigate whether correlations in the input data determine the geometric arrangements of latent features. By systematically manipulating correlation structure in training data, we test how these correlations constrain the degrees of freedom in learned representations. We introduce three correlation patterns to vary the constraint strength. Under the *uncorrelated* condition, features are sampled independently as in the previous subsection. For *paired* correlations each feature in class i is coupled with a feature in class j , such that when one activates, so does its pair. For *global* correlations, each sample has a random phase that determines a cyclic activation pattern: classes adjacent in index have similar activation probabilities and thus frequently co-activate, while classes further separated rarely activate together.

Fig. 2 displays how these correlations determine the arrangement of $\mathbf{v}^{(j)}$. We quantify geometric similarity by comparing the pairwise cosine similarity matrices between all feature pairs in each model, then measuring the correlation between these matrices across different random seeds (25 seed pairs per condition); higher values indicate more similar geometric arrangements. The three correlation regimes

Table 1: PGD-discovered attacks align almost perfectly with the theoretically optimal interference-exploiting perturbation, while a matched-norm random baseline shows no alignment. Cosine similarity (mean $\pm$ std over 1000 attacks per condition); alignment stays at 0.92–0.97 across a $15\times$ range of k , and every configuration exceeds its random baseline of ≈ 0.00 with $p < 10^{-10}$.

k	m	Cosine Sim. (PGD vs Theory)	Cosine Sim. (Random Baseline)
6	2	0.97 ± 0.02	0.00 ± 0.02
30	10	0.96 ± 0.00	0.00 ± 0.01
90	30	0.92 ± 0.00	0.00 ± 0.00



separate cleanly. For the $m = 2, k = 7$ setup, geometric similarity climbs monotonically from 0.23 ± 0.14 (uncorrelated) to 0.47 ± 0.32 (paired) and 0.88 ± 0.06 (global) (mean $\pm$ std) – a near-fourfold increase from the weakest to the strongest constraint. The pattern matches the intuition: uncorrelated data leaves geometries highly variable across seeds, paired correlations partially pin the arrangements, and global correlations force them to near-identical configurations. All three pairwise differences are significant under two-sample t-tests with Bonferroni correction ($p < 10^{-3}$ for every comparison, $\alpha = 0.05$ corrected for 3 comparisons), and the same monotonic ordering holds across every (k, m) configuration we tested (Sec. C.3).

The mechanism is *interference avoidance*: frequently co-activating features are arranged to minimise mutual interference. Stronger correlations impose more constraints: uncorrelated data allows any arrangement that separates classes, paired correlations fix relative positions of coupled features, and global correlations determine a unique geometry up to rotation.

Finding: Input correlations constrain feature geometry. Stronger correlations reduce geometric degrees of freedom, forcing different initialisations to converge to similar arrangements.

Does shared geometry explain attack transferability? Having established that input correlations induce consistent latent geometries across model initialisations, we now test whether the similar interference patterns explains attack transferability. If attacks are tailored for specific interference patterns, they should transfer successfully only between models with similar geometric arrangements.

Using the same seed pairs as the previous paragraph, we generate adversarial attacks on source models and evaluate their success on target models within each correlation condition (25 transfer measurements per condition), with all models achieving $> 95\%$ clean accuracy so that transfer differences cannot be attributed to weak base classifiers. Transfer rates rise in step with geometric similarity. For the $k = 7, m = 2$ configuration (Fig. 2), globally correlated data yields $94\% \pm 12\%$ transfer, more than five times the $18\% \pm 9\%$ seen under uncorrelated data, with paired correlations landing squarely in between at $32\% \pm 20\%$ – exactly the ordering predicted by the geometric-similarity measurements above.

Each perturbation component amplifies or suppresses features, creating constructive interference that pushes representations across decision boundaries. When the same perturbation is applied to a model with different feature arrangements, features that previously constructively interfered instead cancel out or interfere destructively, causing the attack to fail.

Finding: Attack transferability is governed by shared interference patterns.

Does reducing superposition suppress these attacks? If superposition creates vulnerability through interference, then removing superposition should suppress it – and three complementary experiments confirm this in quantitative detail. First, setting $m = k$ forces the network to represent each class direction $\mathbf{v}^{(j)}$ orthogonally; here any prediction-changing perturbation also changes the ground truth, since moving from class A to class B requires the sum of class B features to genuinely exceed class A's. Under this orthogonal regime we find *zero* successful adversarial examples across all 1000 attempts, at every ϵ tested. Second, isolating superposition from capacity by fixing m and varying k shows robust accuracy falling monotonically as k grows, so vulnerability scales directly with the degree of interference rather than with reduced capacity. Third, constraining a single class's latent vector $\mathbf{v}^{(1)}$ to stay orthogonal to all others during training

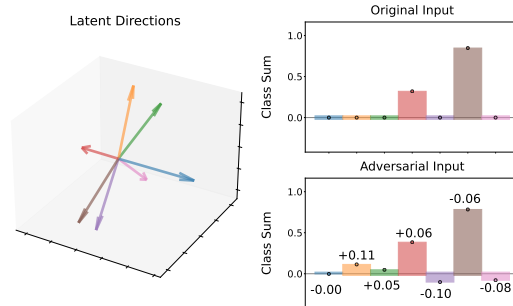


Figure 3: Attacks spend their budget only on superposed features: an adversarial attack (from class 5 to class 3) leaves the input features of the orthogonally represented class (class 1) essentially unperturbed.

localises the effect: attacks between classes still in superposition (e.g. from class a to b with $\mathbf{v}_a \not\perp \mathbf{v}_b$) leave the orthogonal class’s inputs unperturbed (Fig. 3). The attack budget is spent precisely where interference exists and nowhere else.

Finding: Adversarial attacks use their budget to exploit those features in superposition.

3.2 FORMAL ANALYSIS

We now formalise the mechanisms underlying the empirical patterns above. In our linear setting we can derive exact, closed-form relationships between superposition interference and adversarial vulnerability, taking the results beyond observational correlation and giving the near-perfect PGD-to-optimum alignment in Tab. 1 a precise theoretical target. We adopt the notation from our experimental setup, where $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ induces superposition with columns $\mathbf{v}_i$, and the decoder corresponds to the encoder transposed (as empirically observed). The logit for class j is $z_j = \mathbf{v}_j^\top \mathbf{h}$.

Proposition 1. *The optimal input perturbations δ that maximise movement from class j to class k under constraint $\|\delta\|_2 = \epsilon$ satisfy $\delta \propto \mathbf{W}_e^\top \mathbf{n}$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ is the normal to the decision boundary between classes.*

Proof Sketch. To move from class j to k , we maximise the logit margin $z_k - z_j = \mathbf{n}^\top (\mathbf{h} + \Delta \mathbf{h})$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ and $\Delta \mathbf{h} = \mathbf{W}_e \delta$. This becomes $\max_{\|\delta\|_2 = \epsilon} \delta^\top \mathbf{W}_e^\top \mathbf{n}$. By Cauchy-Schwarz, this is maximised when $\delta \propto \mathbf{W}_e^\top \mathbf{n}$. Full proof in Sec. B. $\square$

Corollary 1 (Interference Drives Vulnerability) *The adversarial perturbation magnitude for feature i is $|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$, directly proportional to the differential interference between feature i and the class representations.*

This reveals how superposition creates adversarial vulnerability. Each input feature i is perturbed proportionally to its interference with the class representations. Under superposition, the non-orthogonality means semantically unrelated features interfere with the class decision—adversarial perturbations exploit these dependencies to manipulate outputs.

Proposition 2. *Models with feature representations related by orthogonal transformation $\mathbf{Q}$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$) share identical optimal attack directions in input space.*

Proof Sketch. Under transformation $\mathbf{v}'_i = \mathbf{Q} \mathbf{v}_i$, optimal perturbations remain invariant: $\delta'_i \propto (\mathbf{v}'_i)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = \mathbf{v}_i^\top \mathbf{Q}^\top \mathbf{Q} (\mathbf{v}_k - \mathbf{v}_j) = \mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j) = \delta_i$. Full proof in Sec. B. $\square$

Together with our empirical findings, these propositions establish a **mechanistic pathway**. Specifically: (i) input correlations constrain feature arrangements in superposition, (ii) these geometric arrangements determine interference patterns, (iii) interference patterns dictate optimal perturbations via $\delta \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$ (Proposition 1), and (iv) shared geometric constraints yield similar interference patterns across models, enabling transferability (Proposition 2):

Correlations $\xrightarrow{\text{constrain}}$ **Feature Geometry** $\xrightarrow{\text{determines}}$ **Interference Patterns** $\xrightarrow{\text{enable}}$ **Transferability**

4 ATTACKS IN REALISTIC MODELS

We now extend our analysis to more realistic models, namely a ViT (Dosovitskiy et al., 2020) trained on CIFAR-10 (Krizhevsky, 2009) (and ResNet-18 (He et al., 2016) Sec. D.2.1). By introducing a bottleneck layer directly before the final classification output, we again force the latent class features to be represented in superposition. We then examine: (i) how the degree of superposition impacts attack transferability and robustness, (ii) if a consistent arrangement of the superposed representations emerges across initialisations, and (iii) if the attacks exploit feature interference.

Setup. We train our ViT architecture (6 layers, patch size 4, residual stream $d = 512$) on CIFAR-10 across five random seeds. After training, we replace the original classification head with a bottleneck consisting of a linear encoder that compresses the pre-classification activations to an m -dimensional space, followed by a linear decoder that expands back to the $k = 10$ CIFAR-10 classes. This bottleneck structure, trained with CE loss and no ReLU (mirroring the setup in Sec. 3), forces the k class representations into m -dimensional superposition. During training of the bottleneck layer the

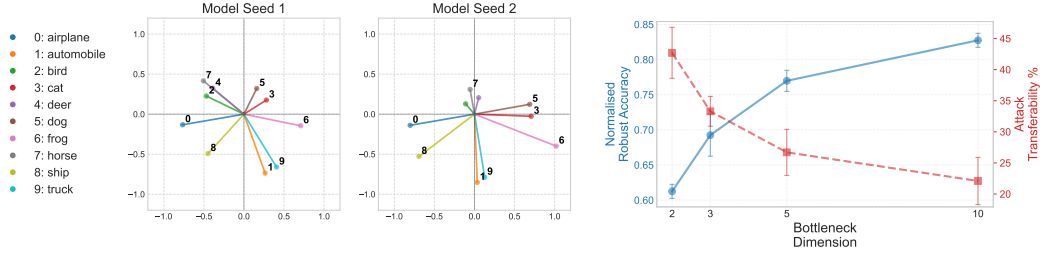


Figure 4: Left: the CIFAR-10 class-representation structure stays consistent across seeds, with the same class groupings recurring between independently trained models. Right: as the bottleneck dimension increases, attack transferability falls and normalised robust accuracy rises – the two curves move in opposite directions in lockstep with superposition pressure, reproducing the toy-model relationship of Sec. 3.

weights of the base ViT are frozen. We vary the bottleneck dimension $m \in \{2, 3, 5, 10\}$ to control the degree of superpositional compression. AEs are generated for each bottlenecked model using both ℓ_∞ - and ℓ_2 -norm PGD. We employ a small step size and a large number of iterations, terminating when an attack successfully changes the model’s prediction (full details in Sec. D.1). Robust accuracy is normalised by the model’s clean accuracy to account for performance variations due to different bottleneck capacities. Attack transferability is measured as defined in Sec. 3.

Results. The base models reach 81% mean clean accuracy, and, as expected, clean accuracy of the bottlenecked models falls smoothly with compression: 81%, 76%, 67%, and 54% at $m = \{10, 5, 3, 2\}$ – a 27-point drop as the latent space is squeezed fivefold, giving us a controlled range of superposition pressures to study.

All three toy-model trends reappear here, confirming that the mechanism is not an artefact of the synthetic setting. First, Fig. 4 (right) shows that tightening the bottleneck m (*i.e.* raising superposition pressure) moves two quantities in opposite, mutually reinforcing directions: (i) normalised robust accuracy *decreases* and (ii) attack transferability across initialisations *increases*. The robustness decline follows from the denser packing of class representations, which lets small input perturbations spread interference across more of the compressed features. The transferability increase follows directly from the Sec. 3 finding that higher superposition removes degrees of freedom from the feature geometry: with fewer viable arrangements available, independently initialised models converge to more similar geometries, share more interference patterns, and therefore transfer attacks more readily. The effect is substantial and monotonic across every ϵ we tested (Sec. D.1.1); for example, under ℓ_∞ PGD at $\epsilon = 0.01$, normalised robust accuracy rises steadily from 61.7% at $m = 2$ to 81.8% at $m = 10$.

Second, and consistent with this, Fig. 4 (left) shows the m -dimensional class representations in the bottleneck settle into a consistent ordering and clustering across seeds – the same shared-geometry signature that drove transferability in Sec. 3, and one that here often tracks semantic similarity (*e.g.* ‘cat’ and ‘dog’ clustering together). Third, when AEs generated for a bottlenecked model are replayed through the corresponding base ViT, they induce closely matching relative logit changes ($\text{logit}_{\text{base}}(\mathbf{x}_{\text{adv}}) - \text{logit}_{\text{base}}(\mathbf{x}_{\text{orig}})$), so both models respond to the same perturbations in the same direction. The perturbations are therefore not merely an artefact of the bottleneck; the added compression is what converts these shared responses into outright misclassifications.

5 ADVERSARIAL VULNERABILITY VIA ALGORITHMIC BRITTLNESS

Prior sections argued that feature superposition can *suffice* for adversarial vulnerability. Here we investigate whether it is *necessary* by studying a setting where features are represented orthogonally, not in superposition, yet remain vulnerable to attacks. This supports a broader view developed in the paper: the analysis of feature representations can be used in robustness analysis. We ask three questions: *how do such attacks work mechanistically, can this understanding be used to construct attacks directly, without gradients, and does naive robustness certification alter the underlying failure mode or merely increase required attack budgets?*

Setup. We train a linear encoder + MLP on one-hot integers (a, b) to predict $(a + b) \bmod P$. As detailed in Sec. E and Nanda et al. (2023), the model discovers a trigonometric algorithm to solve this task. As part of this learned algorithm¹ the encoder projects inputs onto non-basis-aligned linear representations encoding $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at key frequencies $\omega_k = 2\pi k/P$. Ultimately, the MLP composes $\cos((a + b)\omega_k)$ via sum-angle identities to obtain 100% accuracy. Crucially, these features are represented *orthogonally* (i.e. not in superposition). We generate AExs by relaxing the one-hot constraint and perturbing only input a using PGD, moving inputs off the training manifold.

Attacks exploit the learned key frequencies. Fourier analysis of successful perturbations δ (App. Fig. 6) shows their dominant frequencies coincide exactly with the learned algorithm’s key frequencies ω_k . This frequency specificity has a sharp behavioural signature: attacks transfer to models with different ω_k (e.g. those trained from other seeds) at a rate of exactly 0%. The frequency-aligned perturbations cause misclassification by effectively altering the sinusoidal values read off from the encoder features. The perturbation δ transforms one-hot $\mathbf{x}_a$ into a dense vector $\mathbf{x}'_a$. The introduced contributions from the now non-zero inputs corrupt the sinusoidal values, thereby disrupting the MLP’s subsequent computations and leading to misclassification. As the model’s algorithm uses the constructive interference of each ω_k at the logits, disrupting a single ω_k is insufficient. Successful attacks must find δ that maximally alters multiple frequency components. We leave a detailed intuition of the mechanism to Sec. E.1

Analytic construction of attacks. Using this mechanism, we can construct gradient-free, *informed* perturbations formed from composite sinusoidal waves of matching ω_k . The budget these attacks require is the key evidence. On $0 + 0 \bmod P$, the informed attack succeeds at $\|\delta\|_2 = 0.22$ – only $\approx 1.3\times$ the fully gradient-optimised PGD budget of 0.17, yet $4.4\times$ smaller than the best naive baseline (single-index 0.97) and roughly $9\times$ and $28\times$ smaller than ℓ_∞ noise (2.04) and uniform (6.20) perturbations. That a hand-built, gradient-free perturbation lands within a third of PGD while every naive alternative is several-fold more expensive is direct evidence that we have identified the correct mechanism, and shows a representation-aware view of vulnerability yields concrete attack recipes rather than merely post-hoc explanations.

Do defences mitigate this algorithmic vulnerability? We apply certified training against ℓ_∞ perturbations (Müller et al., 2023). At fixed attack budgets, certified training delivers large robust-accuracy gains: from $0\% \rightarrow 85\%$ at $\epsilon_\infty = 5e-3$, $0\% \rightarrow 96\%$ at $2e-3$, and $76\% \rightarrow 100\%$ at $1e-3$ – an improvement of up to 96 points where the undefended model was completely broken. Yet these headline gains are misleading about the underlying failure mode: re-attacked with PGD, the certified model still fails through the *same* frequency-based brittleness, only now at larger perturbation norms. Certification thus raises the price of an attack without removing the specific fragility, which motivates defences that explicitly target the identified semantics (here, frequency manipulation) rather than relying on broad norm bounds alone.

Relation to the paper’s thesis. This section makes two points that integrate with the paper’s broader narrative. First, superposition is *sufficient but not necessary*: orthogonal representations can still be vulnerable to attack. Secondly, viewing AExs through the lens of *architecture–semantics interaction* is actionable: it can guide both the design of attacks (via representation-informed perturbations) and the development of defences.

Although less central here, we note a shared dependence on sparsity assumptions: upstream violations of sparsity (e.g. noise or superposed features) can deliver effectively ‘pre-perturbed’ inputs to downstream orthogonal circuits, activating the same frequency-based failure.

6 RELATED WORK

Superposition & latent representations. Superposition research identifies feature interference mechanisms: Nanda (2023) distinguish *representational* and *computational* superposition; Gurnee et al. (2023) differentiate *alternating* and *simultaneous* interference. Data correlations shape arrangements: pairs of correlated features become orthogonal (Elhage et al., 2022), drive superposition formation (Chan, 2024), and create semantic clustering (Prieto et al., 2025a). Beyond superposition, representation geometry is shaped by: spectral bias (Rahaman et al., 2019), neural collapse (Kothapalli, 2023),

¹We refer to the specific computation executed by the network to solve the task as the *model’s learned algorithm*. As this algorithm depends on extracting precise values from learned feature representations (frequencies), it remains *brittle* in the face of small perturbations to the bases. We term this the *algorithmic brittleness*.

and optimisation objectives (Casper, 2023). Sparse autoencoders (SAEs) have been applied to vision models to disentangle features in superposition (Lim et al., 2024; Joseph et al., 2025).

Adversarial vulnerability. The literature follows an attack-defence dichotomy. Attacks (Szegedy et al., 2014) use imperceptible perturbations to maximize model error, including FGSM (Goodfellow et al., 2014) and PGD (Madry et al., 2018). Adversarial robustness describes maintaining performance under attacks. Defenses include adversarial training (Madry et al., 2018) and certified training (Wong & Kolter, 2018; Palma et al., 2024) providing guarantees against perturbations within input bounds.

The role of input & latent features on adversarial robustness. Vulnerability is attributed to predictive but brittle ‘non-robust’ input features (Ilyas et al., 2019; Goh, 2019), or learning shortcuts (Li et al., 2023). Perturbation characteristics vary by dataset (Maiya et al., 2021). Elhage et al. (2022) suggest that superposition may link to AExs, though Casper (2023) debates this. A tension between representational efficiency and adversarial robustness has been observed (Barsbey et al., 2025). Zhang et al. (2021) show adversarial perturbations push latent representations across decision boundaries. Ganeshan et al. (2019) find PGD targets final layers and propose feature-based attacks. Transferability is limited by representation discrepancies between models (Li et al., 2023; Wang et al., 2024); decorrelating features reduces transfer (Wiedeman & Wang, 2022). Our measurements make this quantitative and directional: as input correlations weaken and feature geometries diverge, transfer drops from 94% to 18%, precisely the trend these works report.

7 DISCUSSION & CONCLUDING REMARKS

We demonstrate that adversarial attacks can exploit interference patterns arising from the geometry of superposed features in NNs, and we back this with a consistent quantitative record. Across toy models and ViTs, data properties – namely correlations and sparsity – induce distinct superposition geometries that create predictable, measurable vulnerabilities: PGD attacks match the interference-exploiting optimum at 0.92–0.97 cosine similarity (versus 0.00 for random baselines), transfer rates track geometric similarity from 18% to 94%, and compressing a CIFAR-10 ViT bottleneck reproduces both the monotonic robustness decline and the transferability rise. These same geometric arrangements let us predict attack transferability and class-specific susceptibility rather than merely observe them. Taken together, the results frame adversarial vulnerability as a potential, inherent consequence of how networks efficiently encode vast amounts of information via superposition.

Limitations & future work. Our quantitative findings stem primarily from simplified and engineered-bottleneck models. While the consistency of the trends across toy models, ViTs, and (in the appendix) ResNet-18 offers an initial mechanistic lens for attacks in large models, the vast adversarial robustness literature suggests that precise vulnerability mechanisms likely vary by attack and model type. For example, constrained attack types (*e.g.* universal perturbations, jailbreaks, patch attacks) may be unable to activate the specific feature combinations required to exploit superposition interference.

Future work includes quantifying the ‘adversarial cost’ of superposition, its connection to accuracy-robustness trade-offs, and how robust training reshapes superposition geometry. Our insights derive from simplified settings with class features in engineered superposition, while real networks involve interference between unknown, unlabelled features across multiple layers. We plan to use SAEs (Bricken et al., 2023) to tackle these challenges in large scale networks, with preliminary results in Sec. F indicating this is a promising direction.

ETHICS STATEMENT

Due to its investigative nature and generality, our mechanistic interpretability framework is not anticipated to have any specific negative impact or immediate societal consequences. Our experiments utilise only publicly available datasets (CIFAR-10) and synthetic toy models, involving no human subjects or personally identifiable information. The gradient-based attack methods used in the paper are already well-established in the literature. Our work does not introduce novel attack capabilities but rather provides theoretical understanding of existing phenomena. We have no conflicts of interest to declare. By elucidating the relationship between superposition and adversarial vulnerability, we hope to inform future research directions that can balance model capability with robustness, ultimately contributing to more trustworthy AI systems that can be safely deployed in critical applications.

REPRODUCIBILITY STATEMENT

We include source code for all experiments in the supplementary materials. The toy model experiments (Sec. 3) include complete architectural details, training procedures, evaluation metrics, and data generation processes. For ViT experiments (Sec. 4), architecture and training procedures are provided, with further details in Sec. D.1. The modular arithmetic experiments (Sec. 5) detail the network architecture, data encoding, and learned trigonometric algorithm, with additional details in Sec. E.2. Experiments were conducted on an AMD Ryzen Threadripper 3970X 32-Core with 256GB RAM and RTX 3090. Additional ResNet-18 and SAE experiments are documented in Appendices Sec. D.2.1 and Sec. F, respectively.

LLM USAGE STATEMENT

This work utilised LLMs to assist with prior literature search, automate LaTeX table generation from CSV files (all numerical values were verified), and suggest alternative phrasings to improve manuscript clarity. All substantive research contributions and conclusions remain the original work of the authors.

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APPENDIX

A GENERATION OF ADVERSARIAL EXAMPLES

We use PGD to generate AExs, which is an iterative method to generate attacks (Madry et al., 2018). For untargeted attacks, it maximises the classifier’s loss function $\mathcal{L}(f(\mathbf{x}'), y_{\text{true}})$; for targeted, it minimises $\mathcal{L}(f(\mathbf{x}'), y_{\text{target}})$ by the update rule:

$$\mathbf{x}'^{(k+1)} = \Pi_{\mathcal{S}} \left(\mathbf{x}'^{(k)} \pm \alpha \mathbf{g}_k \right) \quad (1)$$

(+ for maximisation, – for minimisation). $\mathbf{x}'^{(0)}$ is an initial perturbed input, α is step size, $\mathbf{g}_k$ is the (normalised) gradient $\nabla_{\mathbf{x}} \mathcal{L}(f(\mathbf{x}'^{(k)}), y_{\text{class}})$ (y_{class} being y_{true} or y_{target}), and $\Pi_{\mathcal{S}}$ projects onto the ϵ -ball $\mathcal{S} = \{\mathbf{x}' \mid \|\mathbf{x}' - \mathbf{x}\|_p \leq \epsilon\}$, with input domain clipping. We use the ℓ_p norms:

1. ℓ_{∞} : Constraint $\|\delta\|_{\ell_{\infty}} = \max_i |\delta_i| \leq \epsilon$. $\mathbf{g}_k$ is $\text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(\cdot))$. $\Pi_{\mathcal{S}}$ clips each $\mathbf{x}'_i$ to $[\mathbf{x}_i - \epsilon, \mathbf{x}_i + \epsilon]$.
2. ℓ_2 : Constraint $\|\delta\|_{\ell_2} = \sqrt{\sum_i \delta_i^2} \leq \epsilon$. $\mathbf{g}_k$ is $\nabla_{\mathbf{x}} \mathcal{L}(\cdot) / \|\nabla_{\mathbf{x}} \mathcal{L}(\cdot)\|_{\ell_2}$. $\Pi_{\mathcal{S}}$ rescales $\delta = \mathbf{x}' - \mathbf{x}$ if $\|\delta\|_{\ell_2} > \epsilon$ via $\delta \leftarrow \epsilon \cdot \delta / \|\delta\|_{\ell_2}$.

B OMITTED PROOFS & THEORETICAL RESULTS

To understand how adversarial attacks exploit feature representations, we prove that optimal perturbations weight each input dimension by how much its corresponding feature aligns with the path to the decision boundary. We analyse our linear models without ReLU activation functions.

Consider input $\mathbf{x} \in \mathbb{R}^d$ encoded via $\phi(\mathbf{x}) = \mathbf{W}_e \mathbf{x}$ to latent representation $\mathbf{h} \in \mathbb{R}^m$, where the columns of $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ are an overcomplete dictionary $\{\mathbf{v}_i\}_{i=1}^d$ ($m < d$). For our argmax task, we empirically observe that the trained encoder-decoder pair has the decoder as the transpose of the encoder: $\mathbf{W}_d = \mathbf{W}_e^{\top}$. This means the logit for class j is computed as $z_j = \mathbf{v}_j^{\top} \mathbf{h}$, where $\mathbf{v}_j$ is the j -th column of $\mathbf{W}_e$. The predicted class is $\hat{y} = \arg \max_j z_j$. Input perturbations map to latent perturbations as $\Delta \mathbf{h} = \mathbf{W}_e \delta$, where $\delta = (\delta_1, \dots, \delta_d)^{\top}$ are the perturbation coefficients.

The decision boundary between classes j and k is the set of points where their logits are equal: $\mathcal{B}_{jk} = \{\mathbf{h} : z_j = z_k\}$, or equivalently $\{\mathbf{h} : (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{h} = 0\}$. The vector $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ acts as the normal to this pairwise decision boundary, pointing in the direction that maximally increases the margin $z_k - z_j$. We briefly recall the propositions before providing corresponding proofs.

Proposition 1 (Optimal targeted attack). The optimal input perturbations δ that maximise movement from class j toward class k under constraint $\|\delta\|_2 = \epsilon$ satisfy:

$$\delta \propto \mathbf{W}_e^{\top} \mathbf{n}$$

Proof. To move a sample from being classified as j to being classified as k , we need to maximise the logit margin $z_k - z_j$. Under perturbation δ , the new margin becomes:

$$z'_k - z'_j = \mathbf{v}_k^{\top} (\mathbf{h} + \Delta \mathbf{h}) - \mathbf{v}_j^{\top} (\mathbf{h} + \Delta \mathbf{h}) \quad (2)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^{\top} (\mathbf{h} + \mathbf{W}_e \delta) \quad (3)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{h} + (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{W}_e \delta \quad (4)$$

The change in margin is $(\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{W}_e \delta = \delta^{\top} \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)$. We seek:

$$\max_{\delta} \delta^{\top} \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j) \quad \text{subject to} \quad \|\delta\|_2 = \epsilon$$

Let $\mathbf{g} = \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)$. By the Cauchy-Schwarz inequality:

$$|\delta^{\top} \mathbf{g}| \leq \|\delta\|_2 \|\mathbf{g}\|_2 = \epsilon \|\mathbf{g}\|_2$$

Equality is achieved when δ and $\mathbf{g}$ are parallel. Given the constraint $\|\delta\|_2 = \epsilon$:

$$\delta = \frac{\epsilon}{\|\mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)\|_2} \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)$$

Corollary (Interference drives vulnerability). For a targeted attack from class j to class k , the adversarial perturbation magnitude for input feature i is:

$$|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$$

where $\mathbf{v}_i$ is the i -th column of $\mathbf{W}_e$. This quantity represents the differential interference between feature i and the class representations.

This reveals the mechanism by which superposition creates adversarial vulnerability. Each input feature i is perturbed proportionally to $\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)$ —the differential interference between feature i and the class representations. Under superposition, the non-orthogonality means that even semantically unrelated features have non-zero inner products with $(\mathbf{v}_k - \mathbf{v}_j)$, creating exploitable interference patterns. The multi-class setting amplifies this vulnerability, as with k classes there are multiple possible pairwise boundaries, each creating a distinct interference pattern. Adversarial perturbations leverage these cross-feature dependencies—they manipulate features that affect it through their interference with the class representations. This explains the vulnerability we observe empirically: attacks succeed not by directly increasing target class features, but by exploiting the web of interference created by superposition.

Prop. 1 characterises the optimal perturbation direction for moving from class j to k , providing the gradient for maximising the margin $z_k - z_j$. When intervening classes exist (where $z_j < z_i < z_k$ for some class i), following this gradient might cause the model to predict i before reaching target k . Iterative methods like PGD handle this by recomputing gradients at each step—the global attack path emerges from repeated local decisions rather than a single optimisation. Furthermore, while our analysis assumes $\mathbf{W}_d = \mathbf{W}_e^\top$ based on our empirical observations, the framework extends to arbitrary decoders. In the general case, with decoder $\mathbf{W}_d \in \mathbb{R}^{k \times m}$ having rows $\mathbf{w}_j^\top$, the optimal perturbation becomes $\delta \propto \mathbf{W}_e^\top (\mathbf{w}_k - \mathbf{w}_j)$, capturing interference between encoder directions and decoder weights.

Proposition 2 (Attack transferability). Consider encoders ϕ and ψ with basis matrices $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ and $\mathbf{W}'_e \in \mathbb{R}^{m \times d}$ whose columns are related by orthogonal transformation $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$). If both models use their encoder transpose as decoder (i.e., $\mathbf{W}_d = \mathbf{W}_e^\top$ and $\mathbf{W}'_d = (\mathbf{W}'_e)^\top$), then both models have identical optimal input perturbation vectors for any targeted attack from class j to class k .

Proof. For model ϕ , the optimal perturbation from Prop. 1 is:

$$\delta^\phi \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$$

For model ψ with transformed columns $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$:

$$\delta^\psi \propto (\mathbf{W}'_e)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = (\mathbf{W}'_e)^\top (\mathbf{Q}\mathbf{v}_k - \mathbf{Q}\mathbf{v}_j)$$

Since $\mathbf{W}'_e = \mathbf{Q}\mathbf{W}_e$ (all columns are transformed), we have:

$$\delta^\psi \propto (\mathbf{Q}\mathbf{W}_e)^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{5}$$

$$= \mathbf{W}_e^\top \mathbf{Q}^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{6}$$

$$= \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j) \tag{7}$$

where the last equality uses $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$. Thus δ^ϕ and δ^ψ have identical proportionality. Under the same norm constraint $\|\delta\|_2 = \epsilon$, we have $\delta^\phi = \delta^\psi$. $\square$

Prop. 2 is used to explain why attacks transfer between models with similar training regimes. When models learn feature representations that differ only by orthogonal transformation—essentially the same geometric structure in different orientations—they share identical vulnerability patterns in input space. The orthogonal transformation preserves all inner products between features, maintaining the interference patterns that attacks exploit. This provides an explanation for our empirical observation that models trained on data with the same correlations exhibit high attack transferability: they discover similar feature geometries up to rotation, leading to shared adversarial vulnerabilities.

C TOY MODEL EXPERIMENTS

This section provides supplementary details, extended results and further discussion for the toy model experiments discussed in the main paper. We present model accuracies across a wider range of parameters than shown in the main text, offering insight into how model capacity and data characteristics like sparsity influence the learning process and the conditions under which feature superposition appears. Subsequently, we offer additional visual examples that correspond to Fig. 1, illustrating the mechanics of adversarial attacks under various conditions.

C.1 HYPOTHESES TESTING FRAMEWORK

We explicitly state our hypotheses for the three research questions in Sec. 3.

Research Q1: Do adversarial perturbations exploit superposition geometry?

- H_0 : Adversarial perturbations are random with respect to feature geometry.
- H_1 : Adversarial perturbations systematically exploit geometric relationships between superposed representations.

Research Q2: Do data correlations determine superposition geometry?

- H_0 : Input correlations have no systematic effect on learned geometries.
- H_1 : Input correlations determine geometric arrangements across model initializations.

Research Q3: Does shared geometry explain attack transferability?

- H_0 : Attack transferability is independent of geometric similarity.
- H_1 : Transferability increases with shared latent structure.

We test these hypotheses through controlled experiments:

- $H_1(1)$: We measure the input perturbation profile alignment with a class’s latent representation and the latent attack vector.
- $H_1(2)$: We systematically vary input correlations and measure resulting geometries.
- $H_1(3)$: We quantify transferability rates across models with varying geometric similarity.

C.2 ACCURACY OF TOY MODEL FOR A RANGE OF PARAMETERS

The toy model experiments presented in the main paper predominantly used low-dimensionality settings for conceptual clarity. To demonstrate the model’s behaviour more broadly, this subsection details the classification accuracies achieved by the CE toy model. These results are presented across varying hidden layer size (h), number of classes (k), number of features, and levels of sparsity (S), to provide insight into when the models learn to represent features in superposition. The sparsity level represents the probability that any individual input feature $x_i^{(j)}$ is set to zero, with higher values of S indicating greater input sparsity. We report results as a function of feature density $1 - S$, the probability that a feature is non-zero. Tab. 2 and Tab. 3 provide context on the model’s performance limits and its ability to learn latent representations in superposition.

C.3 CORRELATIONS CONSTRAIN ARRANGEMENTS

Tab. 4 displays the results for the experiments regarding *Do Input Correlations Determine Latent Feature Geometry?* across a wider range of classes (k) and hidden dimension size (m).

C.4 HOW THE TOY MODEL MAPS TO REAL-WORLD MODELS

How does scaling networks affect adversarial vulnerability? While empirical evidence suggests that larger models tend to be more robust to adversarial attacks, this effect is weak. When adversarial training is employed, clearer scaling trends emerge, but improvements remain largely specific to the attack type used during training rather than conferring general robustness (Howe et al., 2025).

Table 2: Classification accuracy of the CE toy model at a fixed bottleneck dimension ($m = 2$), swept over number of classes (k), total input features (features = $k \times 3$), and input feature density ($1 - S$). Reading across each row shows that increasing sparsity (lower density) sharply recovers accuracy – *e.g.* for $k = 10$, accuracy climbs from 0.22 at full density to 0.95 at density 0.02 – while reading down each column shows accuracy falling as more classes are packed into the same two dimensions, quantifying the trade-off between superposition load and sparsity.

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
3	9	2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
4	12	2	0.67	0.71	0.82	0.92	0.98	0.99	1.00	1.00
5	15	2	0.53	0.50	0.65	0.77	0.89	0.95	0.98	0.99
6	18	2	0.40	0.43	0.51	0.66	0.82	0.93	0.97	0.99
7	21	2	0.34	0.34	0.40	0.53	0.73	0.87	0.95	0.98
8	24	2	0.30	0.30	0.33	0.40	0.63	0.82	0.93	0.97
9	27	2	0.24	0.26	0.30	0.35	0.57	0.75	0.89	0.96
10	30	2	0.22	0.24	0.26	0.31	0.50	0.72	0.87	0.95
15	45	2	0.14	0.15	0.16	0.17	0.25	0.40	0.65	0.86
20	60	2	0.10	0.10	0.11	0.13	0.15	0.24	0.44	0.76
25	75	2	0.07	0.08	0.09	0.10	0.12	0.16	0.26	0.62
30	90	2	0.06	0.07	0.07	0.07	0.09	0.11	0.21	0.45

Table 3: Classification accuracy of the CE toy model across numbers of classes (k), total input features, bottleneck dimensions (m), and input feature density ($1 - S$), for varying features per class (p). As in Tab. 2, higher sparsity consistently rescues accuracy at every configuration, confirming that the sparsity–superposition trade-off persists across a broad range of model sizes rather than being specific to the small settings used in the main text.

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
30	30	30	0.23	0.24	0.38	0.62	0.83	0.94	0.99	1.00
30	90	90	0.27	0.27	0.33	0.41	0.51	0.67	0.85	0.94
40	40	30	0.67	0.54	0.73	0.77	0.88	0.96	0.99	0.99
40	120	30	0.71	0.64	0.65	0.72	0.73	0.79	0.89	0.96
60	60	10	0.05	0.07	0.12	0.25	0.47	0.73	0.90	0.97
60	180	10	0.08	0.10	0.13	0.17	0.22	0.32	0.52	0.75
80	80	30	0.15	0.17	0.23	0.41	0.63	0.79	0.91	0.98
80	240	30	0.23	0.22	0.31	0.41	0.48	0.53	0.66	0.80
100	100	10	0.03	0.04	0.05	0.10	0.21	0.43	0.69	0.87
100	300	10	0.04	0.05	0.06	0.09	0.12	0.15	0.25	0.45

Table 4: Geometric similarity across correlation types and superposition configurations. The ordering uncorrelated < paired < global is preserved at every (k, m) : global correlations reach 0.80–0.92 while uncorrelated data stays at 0.17–0.18, a roughly fourfold-to-fivefold gap that holds as the configuration scales from $k=6, m=2$ up to $k=90, m=30$.

Correlation Type	k	m	Geometric Similarity
Uncorrelated	6	2	0.18 ± 0.15
	30	10	0.17 ± 0.02
	90	30	0.17 ± 0.01
Paired	6	2	0.47 ± 0.07
	30	10	0.26 ± 0.07
	90	30	0.25 ± 0.01
Global	6	2	0.92 ± 0.04
	30	10	0.88 ± 0.01
	90	30	0.80 ± 0.01

There are two key phenomena at play when scaling up models. On one hand, larger models have more capacity to represent concepts, but on the other hand, there seems to be a long tail of useful concepts for a larger model to capture in general tasks like next token prediction over internet text. This means that despite increased model capacity, superposition appears to be prevalent even in frontier models (Lindsey et al., 2025). Supporting evidence comes from dictionary learning methods: SAEs require increasingly large dictionaries for larger models (Gao et al., 2025), suggesting that the number of features scales with model size. This suggests that the fundamental tension driving superposition – that models must compress many features into limited dimensions – does not disappear with scale.

Since both superposition and adversarial vulnerability persist in large-scale models, we believe our insights remain relevant across model scales. It is an interesting future avenue to understand how the geometry of superposition changes with scale, potentially helping to mitigate vulnerability.

C.4.1 SEPARATING SUPERPOSITION EFFECTS FROM CAPACITY REDUCTION

Controlling for capacity by keeping the bottleneck dimension fixed to isolate superposition effects controls for the confounding effect of superposition pressure and capacity reduction. We fix bottleneck dimension $m = 2$, and vary number of classes k to isolate superposition pressure. We report the model accuracy, robust accuracy, and transferability in Tab. 5. These controls demonstrate that increased superposition pressure (k/m), independent of model capacity, drives the adversarial vulnerabilities we observe.

C.4.2 ADDITIONAL EXAMPLES OF AEXS IN TOY MODEL

Fig. 1 demonstrates how adversarial attacks exploit the interference between latent features in superposition. Here we provide further visual examples (Fig. 5) to reinforce intuition. Specifically, we supplement the main text by showcasing:

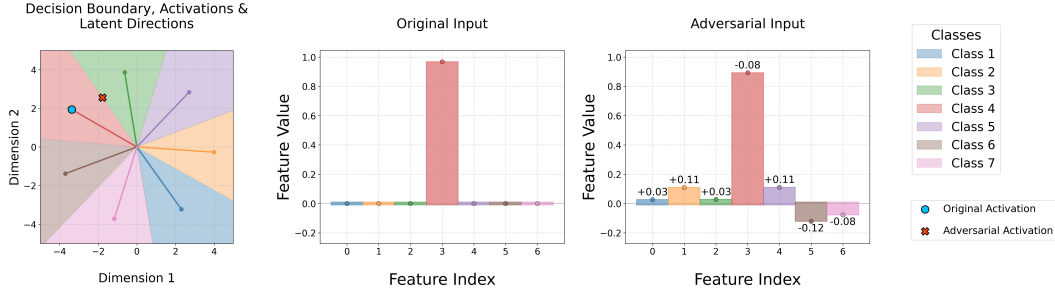
- Fig. 5a shows an additional instance of the setup in Sec. 3.1 ($m = 2, k = 7$) with an ℓ_2 -norm PGD attack, demonstrating the IPP and latent space manipulations that lead to misclassification.
- Fig. 5b shows a similar setup ($m = 2, k = 7$) using an ℓ_∞ -norm PGD attack on a less sparse input.
- Fig. 5c shows an example with increased bottleneck dimensionality ($m = 3, k = 7$) and accompanying ℓ_2 -norm PGD attack. The feature vector similarity matrix used to calculate geometric similarity is also shown.

D CIFAR-10 EXPERIMENTS

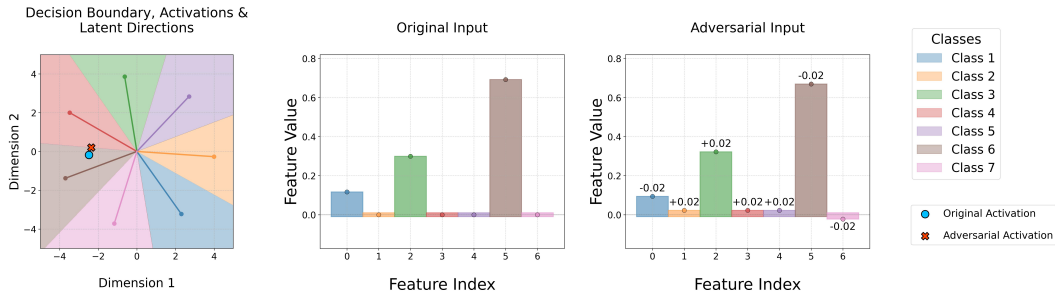
To investigate whether the principles observed in the toy models extend to more complex settings, Sec. 4 introduces experiments trained on CIFAR-10 (Krizhevsky, 2009) with an engineered bottleneck. This appendix section provides further details on this setup and presents extended results.

Table 5: Performance metrics across correlation types, perturbation budgets (ε), and number of classes (k) at fixed bottleneck $m = 2$. Two patterns stand out: robust accuracy falls monotonically as superposition pressure k/m grows (*e.g.* at $\varepsilon = 0.5$, uncorrelated, from 100% at $k = 2$ to 24.9% at $k = 8$), isolating superposition from capacity; and fully correlated data yields dramatically higher transfer (97–100% for $k \geq 4$) than uncorrelated data (0–39%), matching the main-text transferability findings.

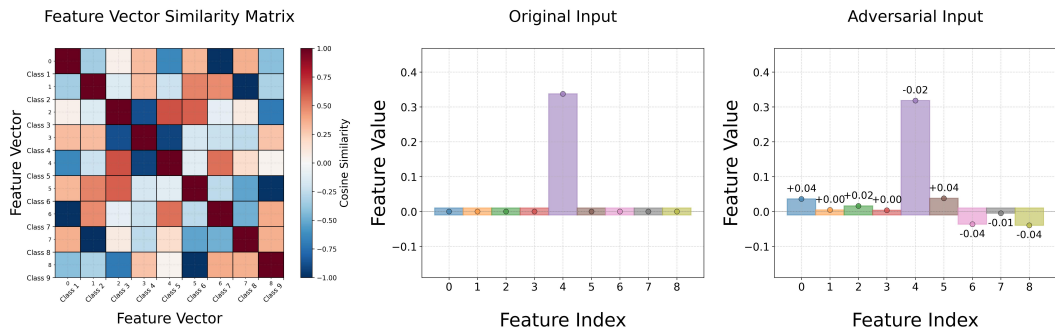
Correlation Type	Parameters		Performance (%)		
	ε	k	Accuracy	Robust Acc.	Attack Trans.
Uncorrelated	0.05	2	100.0	100.0	0.0
		4	100.0	94.9	21.2
		6	99.9	87.0	9.8
		8	99.8	77.7	17.5
	0.1	2	100.0	100.0	0.0
		4	100.0	88.1	9.8
		6	99.7	75.4	9.9
		8	99.6	56.3	23.5
	0.5	2	100.0	100.0	0.0
		4	100.0	99.4	0.0
		6	99.9	62.9	30.5
		8	99.5	24.9	39.0
Fully Correlated	0.05	2	100.0	100.0	0.0
		4	100.0	95.8	98.4
		6	97.3	70.6	98.4
		8	93.0	57.4	97.2
	0.1	2	100.0	100.0	0.0
		4	100.0	88.7	98.5
		6	97.7	45.6	99.7
		8	94.2	34.8	98.6
	0.5	2	100.0	100.0	0.0
		4	100.0	90.0	100.0
		6	97.2	34.6	100.0
		8	93.0	32.0	100.0



(a) An ℓ_2 -norm attack changing the classification of an input of class 4 to class 3. The left plot shows original and adversarial activations in latent space, along with representation directions. The right plots show original and perturbed input feature values.



(b) An ℓ_∞ -norm attack changing the classification of an input of class 6 to class 4. The left plot shows original and adversarial activations in latent space relative to class latent directions. The right plots show original and perturbed input feature values.



(c) An ℓ_2 -norm adversarial attack in a 7-class setup with an increased bottleneck dimension $m = 3$. The leftmost plot now shows the cosine similarity matrix between pairs of latent representations for each of the classes.

Figure 5: Visualisations of AExs in the toy model, supplementing Figure 1 from the main paper by illustrating attack mechanisms in activation space and input space under varied conditions.

D.1 ARCHITECTURE & TRAINING INFORMATION

The base ViT (Dosovitskiy et al., 2020) architecture comprised 6 transformer layers, an embedding dimension (d) of 512, and 8 attention heads in each transformer layer. Input images from the CIFAR-10 dataset, sized at 32×32 pixels, were processed into patches of 4×4 pixels. The Multilayer Perceptron (MLP) within each transformer block had a hidden dimension of 512. Learned positional embeddings were used.

The bottleneck architecture consisted of a linear encoder followed by a linear decoder. The linear encoder projected the pre-classification activations obtained from the ViT backbone into an m -dimensional latent space ($\{2, 3, 5, 10\}$). The subsequent linear decoder then mapped these m -dimensional representations back to the $k = 10$ dimensions corresponding to the CIFAR-10 classes.

The base ViT model was trained on the CIFAR-10 dataset for 250 epochs. A learning rate of 0.001 was used with the Adam optimiser using default PyTorch parameters. The batch size was set to 512 and a cosine annealing learning rate scheduler was used. The loss function was CE. Dropout was used. Training was performed across five different random seeds to account for variability. After the base ViT model was trained, its weights were frozen. The bottleneck layer was then trained for 30 epochs, utilising a learning rate of 0.001. As for preprocessing, the images undergo RandomCrop with 4-pixel padding, resize to the target size (32×32 by default), and RandomHorizontalFlip for data augmentation.

D.1.1 NORMALISED ROBUST ACCURACY ACROSS PERTURBATION MAGNITUDES

Fig. 4 (right) shows how normalised robust accuracy varies with bottleneck dimension for one value of ϵ . This subsection expands on those findings by detailing the normalised robust accuracy when subjected to PGD attacks of different strengths. Results are presented for both ℓ_2 -norm (Tab. 7) and ℓ_∞ -norm (Tab. 6) PGD attacks, providing a more comprehensive view of how the degree of superposition interacts with attack strength to affect model robustness.

AExs for these evaluations were generated using PGD with 100 iterations with a step size (α) of 0.01. Robust accuracy was evaluated on 500 samples for each configuration. The mean normalised robust accuracy and standard deviation across five random seeds are reported.

Table 6: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for ViT models with different bottleneck dimensions (m) on CIFAR-10 under ℓ_∞ -norm PGD attacks of varying magnitude (ϵ), normalised by each bottlenecked model’s clean accuracy. At every attack strength, robust accuracy increases monotonically with m (e.g. at $\epsilon = 0.01$, from 61.7% at $m = 2$ to 81.8% at $m = 10$), so less superposition consistently means more robustness – the trend of Sec. 4 holds across the full ϵ range.

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.001	96.0% $\pm$ 0.4%	97.4% $\pm$ 0.7%	97.9% $\pm$ 0.4%	98.1% $\pm$ 0.3%
0.01	61.7% $\pm$ 3.6%	69.6% $\pm$ 3.5%	77.0% $\pm$ 0.9%	81.8% $\pm$ 3.2%
0.05	4.9% $\pm$ 1.5%	6.7% $\pm$ 1.6%	9.3% $\pm$ 0.6%	10.5% $\pm$ 0.8%
0.1	0.1% $\pm$ 0.2%	0.2% $\pm$ 0.3%	0.2% $\pm$ 0.4%	0.2% $\pm$ 0.4%
0.5	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%

D.1.2 ATTACK TRANSFERABILITY ACROSS PERTURBATION MAGNITUDES

We here include results on attack transferability across various perturbation magnitudes (ϵ) and bottleneck dimensions (m). Tab. 9 presents the ℓ_2 -norm attack transferability and Tab. 8 ℓ_∞ -norm attack transferability.

Table 7: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for ViT models with different bottleneck dimensions (m) on CIFAR-10 under ℓ_2 -norm PGD attacks of varying magnitude (ϵ), normalised by each bottlenecked model’s clean accuracy. The same two monotonic trends as for ℓ_∞ appear: robustness drops with smaller m (more superposition) and with larger ϵ . The m -gap is sizeable at every budget – *e.g.* at $\epsilon = 1.0$, robust accuracy rises from 41.7% ($m = 2$) to 54.9% ($m = 10$) – supporting the findings in Sec. 4.

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.1	90.4% $\pm$ 1.7%	91.8% $\pm$ 0.7%	94.6% $\pm$ 1.2%	95.0% $\pm$ 0.7%
0.5	58.5% $\pm$ 5.0%	60.8% $\pm$ 5.0%	69.2% $\pm$ 1.8%	72.6% $\pm$ 2.8%
1.0	41.7% $\pm$ 4.8%	44.0% $\pm$ 3.0%	50.4% $\pm$ 0.6%	54.9% $\pm$ 1.7%
2.0	34.0% $\pm$ 4.6%	36.2% $\pm$ 3.3%	41.5% $\pm$ 1.3%	47.4% $\pm$ 2.4%
5.0	30.2% $\pm$ 4.1%	32.9% $\pm$ 3.5%	37.3% $\pm$ 1.9%	42.7% $\pm$ 1.0%

D.2 CORRELATIONS IN CLASS FEATURES

In Sec. 3 it was correlations between inputs that drove the superposition arrangements. Here, by contrast, the arrangements are driven not by correlations between input classes but by correlations in the representations at this point in the network – which, at the classification layer, closely track the confusion matrix (*i.e.* which classes are mistaken for which). Classes that cluster together are those the network finds inherently similar and misclassifies together. A controlled comparison confirms this. When we instead fine-tune a stronger base model (timm/vit_base_patch16_384, pretrained on ImageNet-21k – 14 million images, 21,843 classes – and ImageNet – 1 million images, 1,000 classes) on CIFAR-10 and apply the same bottleneck procedure as Sec. 4, the geometry instead shows random ordering across initialisations with approximately equal feature spacing (neural collapse Kothapalli (2023)). This model reaches near-100% accuracy, so its confusion matrix is essentially the identity and the superposed geometry loses its structure. The contrast is the point: models trained solely on CIFAR-10 converge to *similar* geometries precisely because they share the same learned difficulty structure – the same class pairs are hard to separate – whereas a model with no residual confusion has no such shared structure to impose consistent superposition.

D.2.1 RESNET-18

To test architectural generalisation, we repeat the experiments with a ResNet-18 (He et al., 2016) base model in place of the ViT. The base ResNet-18 reaches 92% clean accuracy, 3 points above the ViT’s 89%. At matched ϵ , its normalised robust accuracy falls slightly faster than the ViT’s, but the qualitative signature is identical: robust accuracy still declines and transferability still rises monotonically as bottleneck pressure increases. Both of the core CIFAR-10 trends therefore hold across two very different backbone architectures, indicating the effect is not architecture-specific.

E MODULAR ADDITION

Sec. 5 introduces a modular addition task to investigate adversarial vulnerability in a scenario where features are orthogonal. This section provides a detailed experimental setup for this task, additional results and further discussion on how we view this sort of analysis to be useful.

E.1 LEARNED ALGORITHMIC REPRESENTATION IN MODULAR ADDITION

As detailed in Nanda et al. (2023) and Prieto et al. (2025b), the network discovers and implements a trigonometric algorithm to solve this task: the encoder projects inputs onto non basis-aligned linear representations encoding $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at key frequencies $\omega_k = 2\pi k/P$. These features are represented orthogonally, *i.e.* not in superposition, and the sinusoidal values can be read off via linear probes. The subsequent MLP then applies sum-angle identities to compute $\cos((a + b)\omega_k)$ for various k , which constructively interfere at the logits to select the correct class. AExs are generated

Table 8: Attack transferability (%) for ℓ_∞ -norm PGD attacks on CIFAR-10 ViT models. Transferability is shown from a model trained with a specific ‘Source Seed’ (e.g., Seed 10) to three different target models, each trained with one of the seeds listed in the sub-header (e.g., ‘vs. Seeds 20/30/40’). The three slash-separated values in each cell correspond to the transferability to these three target seeds, respectively. All models within a row share the same bottleneck dimension, m . The ‘Mean $\pm$ Std’ column averages transferability across all 12 source-target seed pairings for each (ϵ, m) configuration. Consistent with Sec. 4, mean transfer generally increases as the bottleneck tightens: at $\epsilon = 0.05$ it rises from 25.4% ($m = 10$) to 38.8% ($m = 2$), and the same ordering recurs at $\epsilon = 0.1$, showing higher superposition yields more consistent latent geometries and thus more transferable attacks.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.001	2	77.8/66.7/44.4	25.0/62.5/25.0	72.7/72.7/72.7	58.3/75.0/58.3	59.3 $\pm$ 17.7
	3	57.1/28.6/28.6	14.3/28.6/28.6	63.6/72.7/72.7	42.9/42.9/42.9	43.6 $\pm$ 18.4
	5	70.0/50.0/50.0	57.1/71.4/57.1	50.0/25.0/25.0	16.7/50.0/33.3	46.3 $\pm$ 16.9
	10	55.6/55.6/66.7	12.5/37.5/25.0	16.7/33.3/16.7	28.6/71.4/28.6	37.3 $\pm$ 19.3
0.01	2	60.3/52.6/48.7	63.8/42.0/60.9	53.8/40.9/52.7	51.2/55.8/46.5	52.4 $\pm$ 6.9
	3	42.9/46.4/50.0	46.4/42.9/49.1	44.2/38.9/44.2	42.9/45.1/41.8	44.6 $\pm$ 3.0
	5	43.3/41.1/43.3	39.2/30.4/41.8	46.0/40.2/47.1	41.2/35.3/32.9	40.2 $\pm$ 4.8
	10	42.5/46.0/44.8	38.0/39.4/46.5	43.5/47.8/47.8	32.6/40.0/35.8	42.1 $\pm$ 4.7
0.05	2	35.0/36.4/37.8	38.8/36.0/46.3	37.4/31.3/41.7	42.0/42.0/40.7	38.8 $\pm$ 3.8
	3	29.9/29.9/37.1	30.4/30.1/35.6	30.2/30.2/35.1	30.1/31.9/34.4	32.1 $\pm$ 2.6
	5	26.8/25.3/29.2	25.2/21.5/26.4	28.0/28.0/26.5	26.0/26.6/21.9	25.9 $\pm$ 2.2
	10	27.1/24.9/30.4	21.6/23.0/28.4	25.3/28.1/30.6	21.0/24.9/20.1	25.4 $\pm$ 3.4
0.1	2	34.2/35.6/38.2	36.7/34.9/44.5	37.9/30.0/40.8	41.2/42.0/40.3	38.0 $\pm$ 3.8
	3	29.6/28.9/35.2	28.6/29.2/34.8	29.1/28.2/34.0	28.8/31.1/32.8	30.9 $\pm$ 2.5
	5	25.4/23.3/27.0	23.8/19.6/24.6	25.9/26.7/25.4	24.6/24.3/20.7	24.3 $\pm$ 2.1
	10	24.8/22.6/29.7	20.1/21.6/26.1	22.5/25.5/28.0	19.5/23.3/18.7	23.5 $\pm$ 3.3

by relaxing the one-hot constraint and perturbing input $\mathbf{x}_j \in \mathbb{R}^P$ (encoding integer j) to $\mathbf{x}' = \mathbf{x} + \delta$ via ℓ_2 and ℓ_∞ -norm PGD. For simplicity, only a is attacked while b is fixed.

As explained in Nanda et al. (2023), the network learns a structured trigonometric algorithm. W_e encodes integers’ sine and cosine values for key frequencies. These features are linearly decodable sine/cosine values. Scaling the corresponding feature vector alters the encoded sinusoidal value itself, not its activation intensity. Integer embeddings $\mathbf{W}_e^j$ project onto a Fourier-like basis, creating a circular/periodic structure in the embedding space. Unlike arrangements from data correlations, this circularity arises from the sine/cosine encoding of the (non-correlated) input integers.

Our model learns to utilise approximately seven key frequencies (ω_k). Each embedding W_e^j for an input integer x_j predominantly encodes its fourteen corresponding trigonometric features: $\cos(\omega_i x_j)$ and $\sin(\omega_i x_j)$ for $i \in \{1, 2, \dots, 7\}$. The concatenated 200-dim. vector thus provides these ≈ 28 key trigonometric features (fourteen for a and fourteen for b) to the MLP. The MLP leverages these, via sum-angle identities (e.g., $\cos(A + B) = \cos A \cos B - \sin A \sin B$), to calculate $(x_1 + x_2) \pmod{P}$. The algorithmic brittleness explored in Section 5 stems from the adversarial perturbation of these key learned trigonometric features.

E.2 EXPERIMENT DETAILS FOR MODULAR ADDITION TASK

The network is trained for addition modulo 113. Inputs are integer pairs (a, b) , $a, b \in \{0, \dots, 112\}$. Each integer is mapped to a 100-dimensional vector via a shared embedding $\mathbf{W}_E \in \mathbb{R}^{113 \times 100}$. Embeddings $\mathbf{W}_E^a$ and $\mathbf{W}_E^b$ are concatenated into a 200-dim. feature vector. This vector feeds a 3-hidden-layer MLP (200 ReLU neurons/layer). The output layer provides 113 logits for the

Table 9: Attack transferability (%) for ℓ_2 -norm PGD attacks on CIFAR-10 ViT models; the source-to-target format mirrors Tab. 8 (see its caption). The ℓ_2 results reproduce the same trend: at $\epsilon = 1.0$, mean transfer climbs from 29.1% ($m = 10$) to 44.6% ($m = 2$), confirming that increased superposition yields more transferable attacks under both norm families.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.1	2	70.6/64.7/41.2	60.0/50.0/60.0	64.3/60.7/57.1	48.1/63.0/51.9	57.6 ± 8.0
	3	39.1/39.1/47.8	50.0/60.7/50.0	48.1/55.6/51.9	48.0/28.0/40.0	46.5 ± 8.4
	5	43.5/34.8/43.5	31.6/31.6/42.1	42.9/35.7/21.4	30.4/39.1/34.8	35.9 ± 6.4
	10	45.5/59.1/45.5	19.0/28.6/28.6	37.5/43.8/43.8	23.8/52.4/38.1	38.8 ± 11.4
0.5	2	59.6/52.1/48.9	60.8/35.4/60.8	54.1/39.4/51.4	43.8/49.5/46.7	50.2 ± 7.7
	3	38.7/40.6/48.1	36.1/36.8/41.7	37.4/36.5/43.5	34.4/34.4/36.7	38.7 ± 3.9
	5	38.4/32.0/37.6	29.4/25.7/32.1	39.1/33.9/38.3	31.8/29.9/32.7	33.4 ± 4.0
	10	38.3/37.4/37.4	26.5/28.4/31.4	32.4/35.3/35.3	23.4/28.2/25.0	31.6 ± 5.0
1.0	2	50.8/43.7/44.4	51.7/36.4/50.0	46.1/32.9/44.7	42.7/46.7/45.3	44.6 ± 5.3
	3	35.7/36.9/41.7	31.8/36.4/42.6	33.7/32.0/37.3	34.9/32.0/35.5	35.9 ± 3.3
	5	32.1/28.9/33.2	30.9/23.0/30.9	35.9/31.5/33.7	27.7/28.2/27.1	30.3 ± 3.3
	10	32.3/32.3/38.5	23.7/27.7/29.9	29.1/32.0/34.3	22.2/25.6/21.1	29.1 ± 5.0
2.0	2	47.0/41.6/44.3	48.2/36.2/48.2	43.9/33.5/42.1	43.4/43.4/42.8	42.9 ± 4.2
	3	31.9/37.7/40.3	31.1/35.6/40.6	32.6/31.1/36.8	35.4/33.3/34.3	35.1 ± 3.2
	5	29.9/25.4/32.1	26.3/22.1/26.7	32.3/31.8/29.5	27.9/27.9/23.6	28.0 ± 3.2
	10	30.8/30.4/35.7	22.5/24.9/28.6	24.8/29.7/32.7	20.4/26.1/19.0	27.1 ± 4.9
5.0	2	44.7/46.7/40.8	45.3/37.3/48.7	42.7/33.9/41.5	42.3/42.3/40.6	42.2 ± 3.8
	3	32.3/35.8/40.8	31.3/32.6/39.9	31.0/30.0/33.8	35.0/33.0/33.0	34.1 ± 3.2
	5	30.6/24.1/28.6	28.3/20.9/24.8	31.7/31.3/27.4	28.1/29.0/23.2	27.3 ± 3.3
	10	29.8/27.4/32.7	19.7/24.1/28.9	24.9/29.9/31.2	18.8/25.8/19.2	26.0 ± 4.6

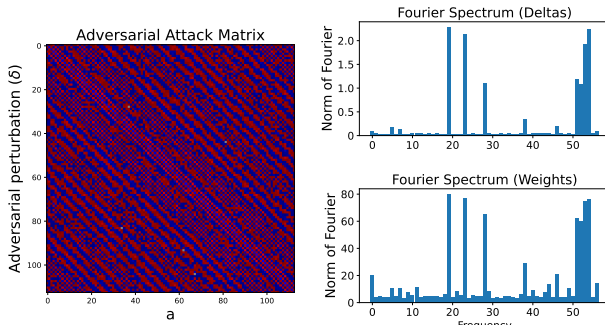


Figure 6: Left: each column is the mean attack δ added to a to successfully attack $a + b \bmod p$. Right: the norms of the Fourier components of δ line up precisely with those of $W_{MLP_{out}}$, showing the perturbation concentrates its energy on exactly the key frequencies the learned algorithm uses.

Table 10: Minimum ℓ_2 -norm budget to successfully attack $0 + 0 \bmod p$, by attack type. The representation-informed attack (0.22) comes within $1.3\times$ of gradient-optimised PGD (0.17) yet needs $4.4\times$ less budget than the best naive baseline (single-index 0.97) and $9\text{--}28\times$ less than ℓ_∞ noise (2.04) and uniform (6.20) perturbations.

Attack Method	Success ϵ
PGD	0.17
Informed Attack	0.22
Single a	0.97
ℓ_∞ Noise	2.04
Uniform δ	6.20

classification. The model is trained end-to-end using cross-entropy loss and AdamW (weight decay 4), achieving 100% test accuracy. AExs are generated using PGD under ℓ_2 and ℓ_∞ norms with

varying ϵ . Attacks perturb the embedding $\mathbf{W}_E^a$ (with integer b fixed) by adding a perturbation δ to cause misclassification of $(a + b) \pmod{113}$.

E.3 ALGORITHMIC BRITTLINESS

The frequency-aligned perturbations cause misclassification by effectively altering the sinusoidal values read off from the encoder features. We use the hypothesised neuron-aligned case to give intuition: $\mathbf{W}_e$ would contain a row corresponding to the sine/cosine for each ω_k , whose element at column $\mathbf{w}_e^a$ contains the value $\sin(\omega_k a)$ or $\cos(\omega_k a)$. For the input a , the one-hot $\mathbf{x}_a$ acts to select $\mathbf{w}_e^a$, giving $\mathbf{h}_a = \mathbf{w}_e^a$. The perturbation δ transforms one-hot $\mathbf{x}_a$ into a dense vector $\mathbf{x}'_a$. Now $\mathbf{h}'_a = \mathbf{W}_e \mathbf{x}'_a = (1 + \epsilon_a) \mathbf{h}_a + \sum_{k \neq a} \epsilon_k \mathbf{w}_e^k$. The introduced contributions from the $k \neq a$ sinusoidal terms corrupts the sinusoidal values, thereby disrupting the MLP’s subsequent computations and leading to misclassification. The model’s algorithm uses the constructive interference of each ω_k at the logits, and as such disrupting a single ω_k is insufficient as the other components overwhelmingly support the correct answer. Thus, effective attacks must find δ that maximally alters multiple frequency components.

E.4 ANALYTIC CONSTRUCTION OF ATTACKS AND BASELINE COMPARISONS

The main paper describes how understanding the learned trigonometric algorithm in the modular addition task allows for the analytic construction of adversarial attacks. Sec. 5 of the main paper benchmarks this ‘informed attack’ against PGD and naive baselines. This section provides further details on these baseline attack methods. The baselines are:

- **PGD Attack:** Standard PGD used to find the adversary.
- **Informed Attack:** The analytically constructed perturbation, from combinations of cosines.
- **Single a :** This involved finding the minimal attack to a single input to cause misclassification.
- **ℓ_∞ Noise:** This baseline involved adding random noise δ_{noise} to the input a . Each component of δ_{noise} was sampled independently and uniformly from $[-\epsilon, \epsilon]$. The ϵ that successfully induced a misclassification is reported.
- **Uniform δ :** The perturbation vector δ is constrained to have all components equal, i.e., $\delta_j = c \cdot 1$, where c is a scalar. The magnitude of c is reported such that adding δ to the input causes a misclassification.

E.5 DETAILS ON CERTIFIED ROBUST TRAINING

To investigate the impact of defences on the identified vulnerability, we performed certified robust training. This aimed to make the network provably robust against ℓ_∞ -norm perturbations. The certified training method employed was an abstraction-based technique (Wong & Kolter, 2018; Raghunathan et al., 2018; Palma et al., 2022), namely RSIP-IBP (Zhang et al., 2020; Henriksen & Lomuscio, 2023). RSIP-IBP utilises Interval Bound Propagation (IBP) (Katz et al., 2017; Ehlers, 2017) and Reversed Symbolic Interval Propagation (RSIP) to compute reachable output sets for network layers, thereby enabling formal certification of robustness within a given perturbation budget. The training loss function incorporated both a standard loss component and an RSIP loss component. Certified training was conducted for 1000 epochs, with the initial 200 epochs serving as a warm-up phase. The learning rate was 1×10^{-3} and weight decay 1×10^{-4} . The same train/test data splits were used as for the standard model training. The results of certified training across different training perturbation budgets (ϵ_{train}) and evaluated against various attack perturbation budgets (ϵ_{attack}) are summarised in Tab. 11.

As discussed in Sec. 5, when these robustly trained models were subsequently attacked using PGD, the nature of successful adversarial perturbations remained consistent: they continued to exploit the same key frequency components inherent to the model’s learned algorithm. The primary effect of the certified training was an increase in the perturbation magnitude required to achieve misclassification for those epsilons against which the model was successfully certified, rather than a fundamental change in the attack vector’s alignment with the model’s algorithmic sensitivities.

Table 11: Verified robust accuracy (%) for modular addition models under ℓ_∞ -norm perturbations, comparing a standard (pre-robustification) model with models certifiably trained via RSIP-IBP at different training budgets (ϵ_{train}), evaluated against various attack budgets (ϵ_{attack}). Certified training raises verified accuracy from 0.0% to 85.3–100% at the budgets it targets (e.g. 0.0% $\rightarrow$ 95.7% at $\epsilon_{\text{attack}} = 2\text{e-}3$ with $\epsilon_{\text{train}} = 10^{-3}$), yet as discussed in Sec. 5 PGD still succeeds at higher ϵ_{attack} by exploiting the same algorithmic sensitivities – the defence raises the price of an attack without removing the mechanism.

Attack Epsilon (ϵ_{attack})	Train Epsilon (ϵ_{train})	Standard Acc. (Pre-Rob., %)	Robust Acc. (Pre-Rob., %)	Standard Acc. (Robust, %)	Robust Acc. (Robust, %)
5.0×10^{-4}	10^{-4}	100.0	100.0	100.0	100.0
	10^{-3}				100.0
1.0×10^{-3}	10^{-4}	100.0	76.0	100.0	93.8
	10^{-3}				100.0
2.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				95.7
5.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				85.3

E.6 ON A POTENTIAL RECONCILIATION WITH SUPERPOSITION VULNERABILITY

Although the vulnerability explored here arises in the absence of feature superposition, it exhibits a notable phenomenological resemblance to those driven by superposition — a shared dependence on sparsity in the input representation. This motivates our *reconciliation conjecture*: algorithmic vulnerabilities may persist in networks with orthogonal representations when sparsity assumptions are violated upstream. If earlier layers in a network produce noisy or non-sparse features, potentially due to adversarial interference, then these deviations can propagate into otherwise well-isolated downstream circuits. In such cases, circuits that operate correctly on clean inputs may nonetheless fail, as the input they receive is effectively already perturbed. This provides a plausible route by which algorithm-disruption mechanisms, like the frequency attacks studied here, could be activated even in the absence of superposition. Further, this interaction between upstream superposition and downstream algorithmic brittleness, and more broadly layer-by-layer understanding of the specific vulnerability mechanisms triggered by AExs, is underexplored. We leave a detailed investigation, grounded in empirical evidence and theoretical modelling, as a promising direction for future work.

F INSPECTING ADVERSARIAL PERTURBATIONS VIA SAE FEATURES

The analyses of the main paper focused on scenarios with either explicitly defined class features (toy model) or class-level representations in a bottleneck (ViT experiments). However, in large, practical networks, interference will occur between an unlabelled latent features across all layers. As discussed in the future work section of the main paper, SAEs offer a promising unsupervised method to extract more linear features from NNs (Bricken et al., 2023). This appendix section presents a preliminary investigation into using SAE-extracted features to characterise how a large scale ViT responds internally when processing AExs versus clean inputs. These initial results are intended to explore the feasibility of this approach for future work aimed at understanding adversarial phenomena on a feature level within large models, rather than presenting a conclusive new set of findings. We first outline the experimental setup and validate the SAEs behaviour on adversarial inputs before presenting exploratory layer-wise feature difference analyses.

F.1 EXPERIMENT SETUP

We analyse SAE-extracted features from the vision encoder of the LAION/CLIP-ViT-B-32-DataComp.XL-s13B-b90K model, a 12-layer ViT with $\approx 70\%$ zero-shot accuracy on ImageNet-1k Deng et al. (2009). We use open-source Prisma SAEs mapping the 768-dimensional activation

space to a dictionary of 49,152 features ($64 \times$ expansion) (Joseph et al., 2025). We examine SAEs trained on either mean patch token representations or CLS token representations, across all layers. We analyse features activating above a high threshold (0.1) to capture changes to the highest activating features, and a low threshold (0.001). We generate 10,000 original/successfully-attacked image pairs (ℓ_∞ -norm, $\epsilon = 0.02$) for our SAE feature comparisons. This setup allows for an initial exploration of how adversarial perturbations might manifest at the level of SAE features.

F.2 SAE VALIDITY FOR ADVERSARIAL INPUTS

Given that AExs are out-of-distribution inputs, and our aim is to use SAE (standardly trained) to understand them, we first conduct a basic validation. This subsection examines the impact on model accuracy and SAE reconstruction when SAEs are inserted at inference time for both clean and adversarial inputs. We are investigating whether the SAEs process features relevant to the classification task, even under adversarial conditions, before attempting to interpret differences in feature activations.

Table 12: Impact on ImageNet-1k top-1 accuracy of inserting a pre-trained SAE (one layer at a time) at inference, for original, ℓ_∞ adversarial ($\epsilon = 0.02$), and random noisy inputs. ‘Original input + SAE’ passes the clean ViT activations at a layer through the SAE and forwards its reconstruction. The SAE preserves nearly all clean performance (0.99 vs 1.00 at layers 2–8) and adds only a small ‘de-attacking’ effect on adversarial inputs (accuracy rising from 0.00 to at most 0.11 at layer 11); the model still misclassifies the overwhelming majority of adversarial inputs, indicating the SAE passes through the salient changes that drive misclassification rather than filtering them out.

Metric	Layer 2	Layer 5	Layer 8	Layer 11
Accuracy				
Original input	1.00	1.00	1.00	1.00
Original input + SAE	0.99	0.99	0.99	0.86
Adv. input	0.00	0.00	0.00	0.00
Adv. input + SAE	0.02	0.01	0.01	0.11
Noisy input	0.98	0.98	0.98	0.98
Noisy input + SAE	0.98	0.98	0.98	0.85
SAE Improvement				
Original input	-0.01	-0.01	-0.01	-0.14
Adv. input	0.02	0.01	0.01	0.11
Noisy input	-0.00	-0.00	-0.00	-0.13

F.3 LAYER-WISE SAE FEATURE DIFFERENCES

This subsection presents initial exploratory visualisations comparing SAE feature activations across different layers of the ViT for original, ℓ_∞ adversarial, and random noisy inputs. These visualisations are intended to highlight potential avenues for future, more in-depth investigation into how adversarial attacks distinctively alter sparse feature representations. We examine:

- **Activation count:** Histograms comparing the number of SAE features activated above a set threshold in a layer for original, adversarial, and noisy images. This explores whether attacks characteristically alter feature sparsity.
- **Feature overlap:** The degree of overlap (number of common activated features) between (original vs. adversarial) and (original vs. noisy) conditions. This explores if attacks activate a distinctly different set of features.
- **Jaccard Similarity:** The Jaccard similarity between sets of activated features, offering a normalised measure of overlap.
- **Mean activation:** Histograms of mean activation values for features active above a threshold. This looks for changes in the intensity of (key) features.
- **Distinct Features:** The count of unique features activated across an image set for each condition.

F.4 CLS TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection presents a preliminary analysis of SAE features derived from CLS tokens, only considering features activated above a threshold of 0.1. Figures 7 through 10 visualise and compare the histogram of activation counts, feature overlap, Jaccard similarity, and histogram of mean activations, respectively, for original, adversarial, and random noisy inputs across different ViT layers.

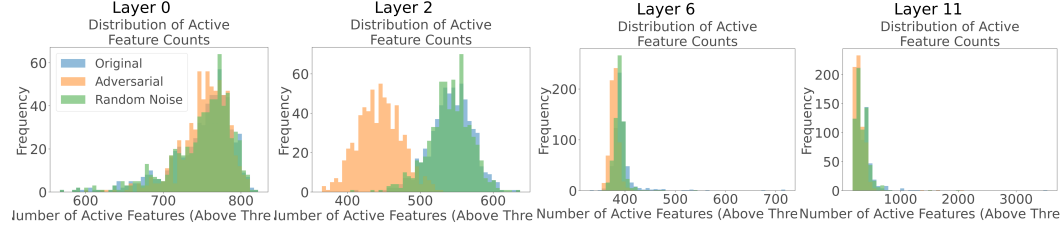


Figure 7: Histogram of activation counts for CLS token SAE features (threshold 0.1).

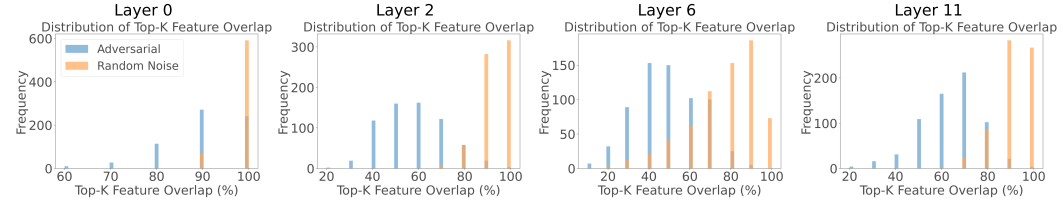


Figure 8: Feature overlap between original vs. attacked and original vs. noisy images for CLS token SAE features (threshold 0.1).

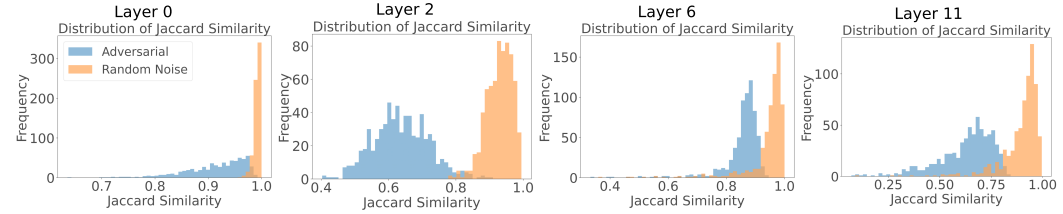


Figure 9: Comparison of Jaccard similarity for CLS token SAE features (threshold 0.1).

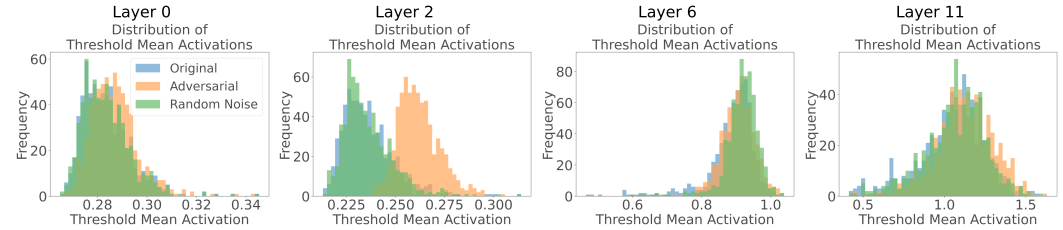


Figure 10: Histogram of mean thresholds for CLS token SAE features (threshold 0.1).

F.5 CLS TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

Here, we examine SAE features from CLS tokens with a lower activation threshold of 0.001, to not just capture the most intensely activated features. The subsequent figures illustrate the histogram of activation counts (Fig. 11), feature overlap (Fig. 12), Jaccard similarity (Fig. 13), and histogram of mean activations (Fig. 14) across layers for original, adversarial, and noisy inputs.

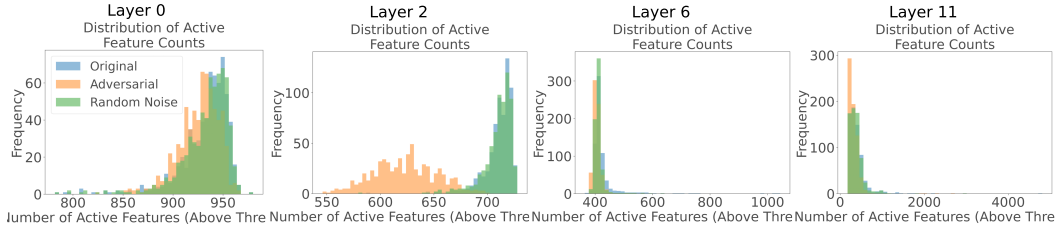


Figure 11: Histogram of activation counts for CLS token SAE features (threshold 0.001), with noise.

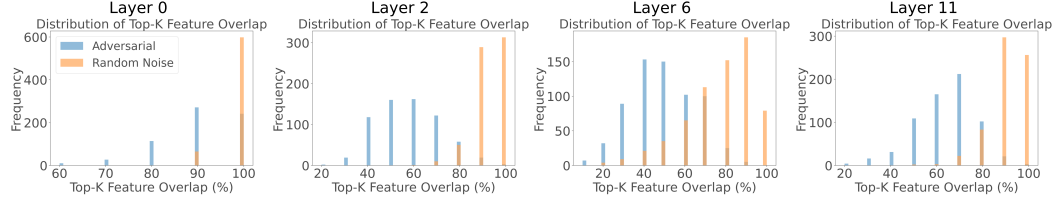


Figure 12: Feature overlap between original vs. attacked and original vs. noisy images for CLS token SAE features (threshold 0.001).

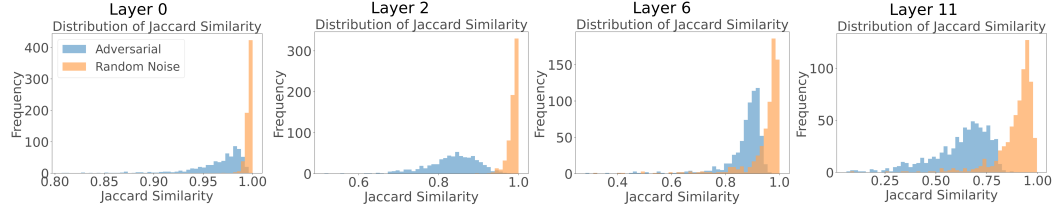


Figure 13: Comparison of Jaccard similarity for CLS token SAE features (threshold 0.001).

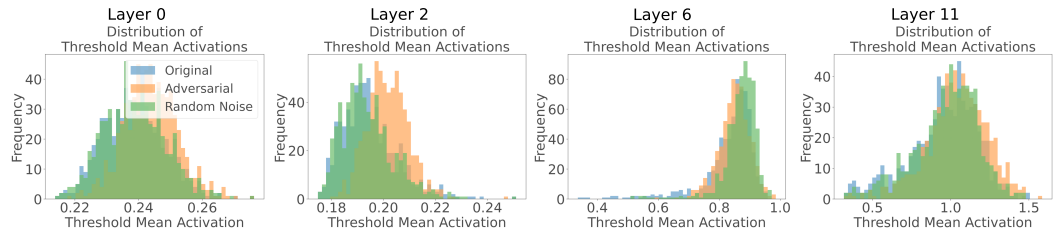


Figure 14: Histogram of mean activations for CLS token SAE features (threshold 0.001), with noise.

F.6 PATCH TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection shifts the focus to SAE features derived from the mean of patch tokens, again using a high activation threshold of 0.1. We present comparisons of activation count histograms (Fig. 15), distinct feature count histograms (Fig. 16), feature overlap (Fig. 17), Jaccard similarity (Fig. 18), and mean activation histograms (Fig. 19) for original, adversarial, and noisy inputs.

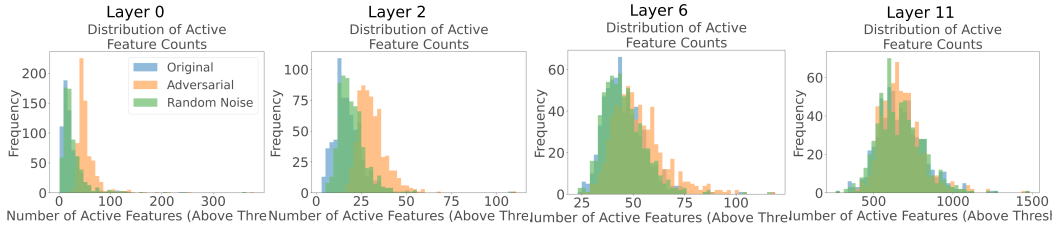


Figure 15: Histogram of activation counts for patch token SAE features (threshold 0.1), with noise.

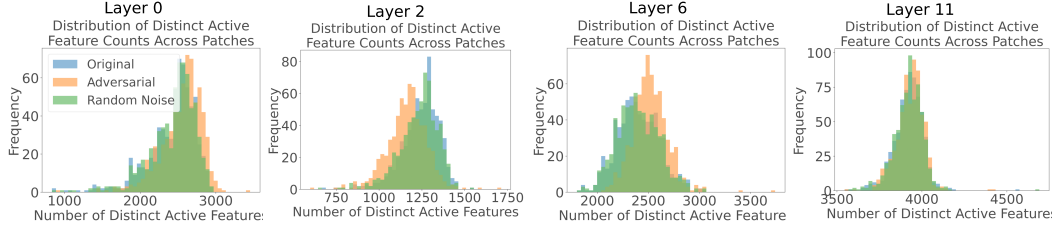


Figure 16: Histogram of distinct patch features for patch token SAE features (threshold 0.1), with noise.

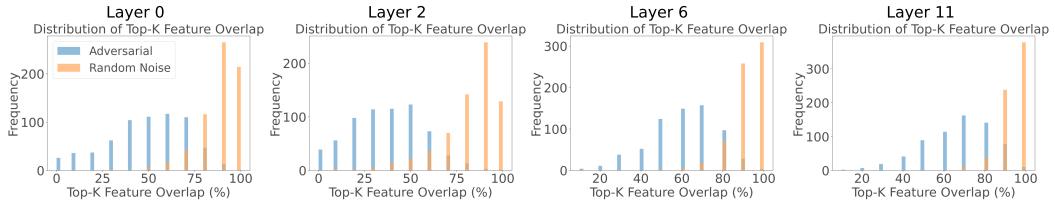


Figure 17: Feature overlap between original vs. attacked and original vs. noisy images for patch token SAE features (threshold 0.1).

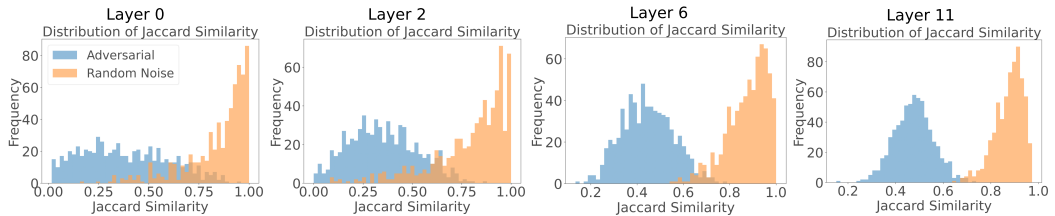


Figure 18: Comparison of Jaccard similarity for patch token SAE features (threshold 0.1).

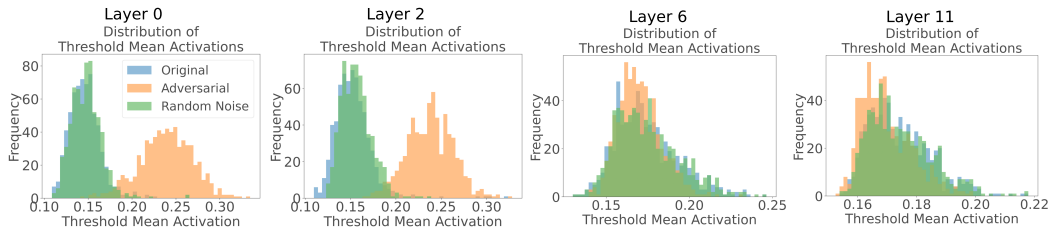


Figure 19: Histogram of mean thresholds for patch token SAE features (threshold 0.1), with noise.

F.7 PATCH TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

Finally, this subsection explores SAE features from the patch tokens means using a low activation threshold of 0.001. The figures show histograms of activation counts (Fig. 20), distinct feature counts (Fig. 21), feature overlap (Fig. 22), Jaccard similarity (Fig. 23), and mean activation histograms (Fig. 24) for original, adversarial, and noisy inputs.

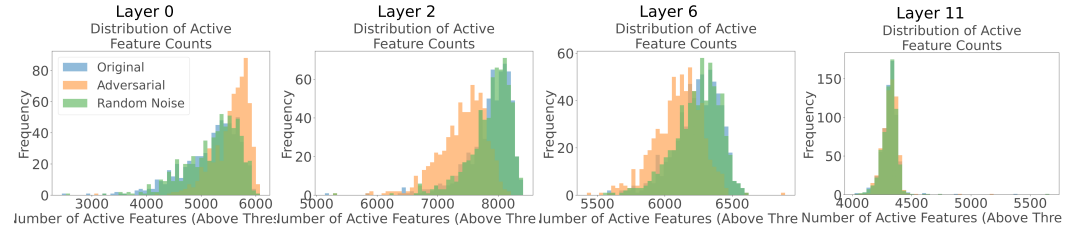


Figure 20: Histogram of activation counts for patch token SAE features (threshold 0.001), with noise.

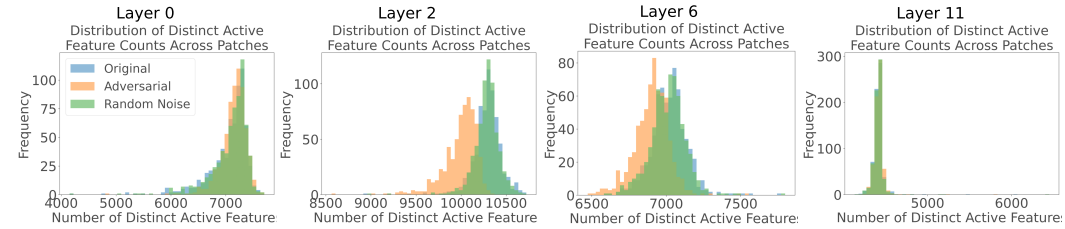


Figure 21: Histogram of distinct patch features for patch token SAE features (threshold 0.001), with noise.

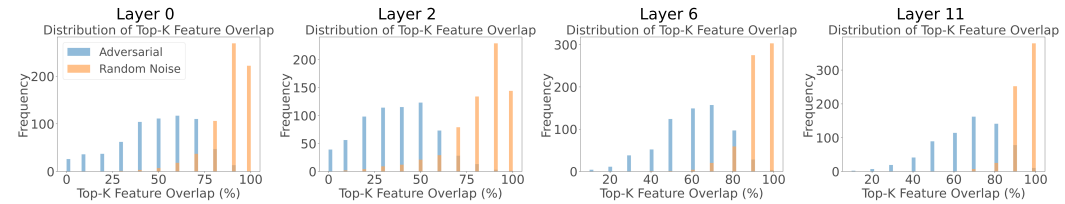


Figure 22: Feature overlap between original vs. attacked and original vs. noisy images for patch token SAE features (threshold 0.001).

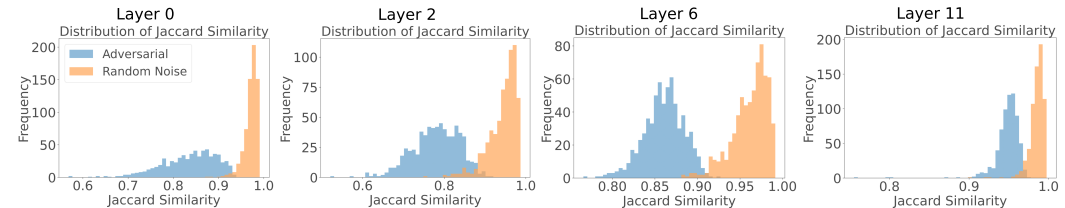


Figure 23: Comparison of Jaccard similarity for patch token SAE features (threshold 0.001).

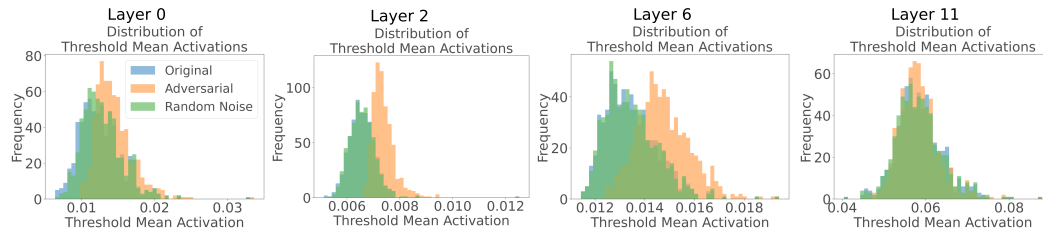


Figure 24: Histogram of mean activations for patch token SAE features (threshold 0.001), with noise.

G ACRONYMS

AEx adversarial example

CE cross entropy

LRH linear representation hypothesis

MLP Multilayer Perceptron

MSE mean squared error

NN neural network

PGD projected gradient descent

SAE sparse autoencoder

VIT vision transformer